

EXHIBIT 34

MPS Powers Every AI Socket

MPS电源方案助力每一个人工智能硬件系统

March 2024

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现场应用工程师经理





Agenda

- MPS Company Introduction;
- MPS Power Solutions for Data Center with AI/HPC;
 - What's Lateral Power Delivery(LPD) and Z-axis Power Delivery(ZPD™), and why ZPD™?
 - MPS LPD and ZPD™ solutions.

Monolithic Power System

MPS — A worldwide silicon-based power solutions company

Headquarters — Kirkland, WA

Founded

1997

in Silicon Valley, IPO in 2004 (NASDAQ: MPWR)

Revenue

\$1.8B

in 2023 (1.5% growth from 2022)

Employees

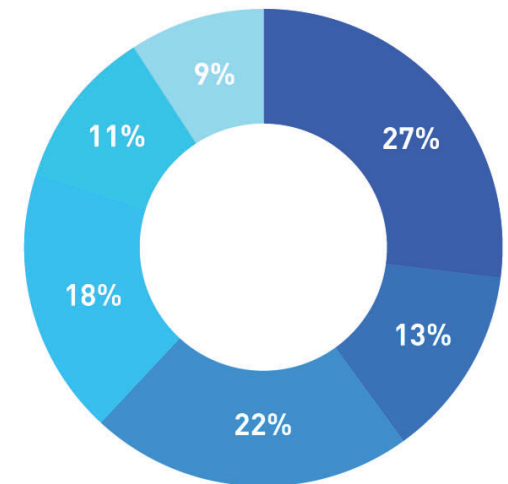
3,500+

Worldwide

Products

4,000+

Products in Cloud Computing, Automotive, Industrial, Consumer Electronics, and More



- Storage and Computing
- Consumer
- Automotive
- Enterprise Data
- Communications
- Industrial/Other

MPS

CORPORATE PROFILE

Production Facilities

Testing

- MPS Chengdu Facility
- MPS Taiwan Facility
- Malaysia & China Taiwan

Assembly

- Malaysia
- China Taiwan
- China Mainland

IATF-Certified Wafer Fabs

- | | |
|---------------------|--------------------------|
| • Key Foundry/Hynix | Korea |
| • SSMC | Singapore |
| • TSMC | China Taiwan |
| • Vanguard | Singapore & China Taiwan |
| • SMIC | China Mainland |
| • HHGrace | China Mainland |
| • ASMC | China Mainland |

Quality & Reliability Facilities

- Both San Jose and Chengdu
- Burn-In Room: 3,000 Sq. Ft.
- Two State-of-the-Art Failure Analysis Labs



EMC Lab Facilities



MPS PRODUCT FAMILIES

AC/DC Power Conversion

- High-Voltage Buck Regulators
- High-Voltage LDOs
- Flyback Controllers
- Flyback Synchronous Rectifiers
- Active PFC Controllers
- LLC Resonant Converter Controllers
- LLC Synchronous Rectifiers
- PFC & LLC Combo Controllers
- X-Capacitor Bleeders

Display Backlighting Power

- Backlight Drivers
- Electroluminescent Drivers
- Photo Flash Drivers
- LCD Power Supplies

AI & Computing Power

- CPU & GPU Core Power
- High-Current DrMOS
 - 90A DrMOS 4mmx6mm Package
- 25A POL Module 5mmx6mm Package
- 60V Intermediate Bus Converter Modules
- Z-Axis Power Delivery Multiphase Modules

DC/DC Power Conversion

- Step-Up (Boost)
- Step-Down (Buck)
- Buck-Boost
- PMIC
- Power Magnetics
 - Molded Inductors
 - Semi-Shielded Inductor
- Power Over Ethernet (POE)
 - Power Sourcing Equipment (PSE)
 - Powered Devices

E-Fuse, USB & Load Switches

- Configurable Current Limit, Up to 50A per Device
- Adjustable Slew Rate
- Reverse Current Blocking
- Output Discharge (Load Switch)
- Integrated Auto-Detection
- Pin-to-Pin Compatible
- Parallelable Up to 10 Devices
- Digital Interface Command and Control
- USB-C and USB PD

Motor Drivers, Position & Current Sensors

- Brushless DC Motor Drivers
- Stepper Motor Drivers
- Brushed DC Motor/Solenoid Drivers
- Half-Bridge/Full-Bridge/Three-Phase Power Stages
- Magnetic Angular Position Sensors
- Current Sensors, 5A-50A Isolated

Automotive & Industrial

- AEC-Q100
 - DC/DC
 - LED Lighting
 - Power Modules
 - Motor Drivers
 - USB Charging
 - Wireless Charging
 - Display Backlighting
 - Precision Analog
 - ADAS Core Power
 - Supervisors
 - Load Switches

Battery Management

- Battery Monitors & Protectors
- Fuel Gauges
- Active Balancers
- Switching Chargers: Buck, Boost, Buck-Boost
- Linear Chargers

Class-D Audio

- Analog Input Class-D Amplifiers
- PWM Input Power Stages

Isolated Solutions

- Isolated Gate Drivers
- Isolated Power Module
- Digital Isolator with Integrated Power
- Digital Isolators
- Totem-Pole PFC/LLC Controller

Power Modules

- 6V, 600mA to 100A
- 16V, 600mA to 1000A+
- 36V, 600mA to 24A
- 55V, 1A to 3A
- 80V, 600mA

LED Lighting & Illumination

- TRIAC-Dimmable AC/DC LED Controllers
- PWM and Analog Dimmable AC/DC LED Controllers
- DC/DC LED Controllers: Buck, Boost, Buck-Boost
- LED Protection ICs
- IR LED Driver

Precision Analog

- Analog-to-Digital Converters
- Analog Switches
- Current-Sense Amplifiers
- Operational Amplifiers
- Voltage Reference

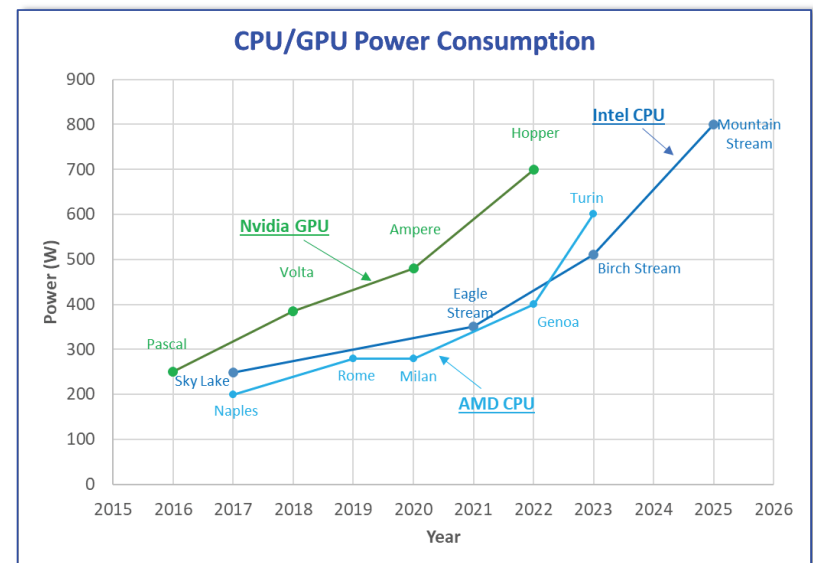
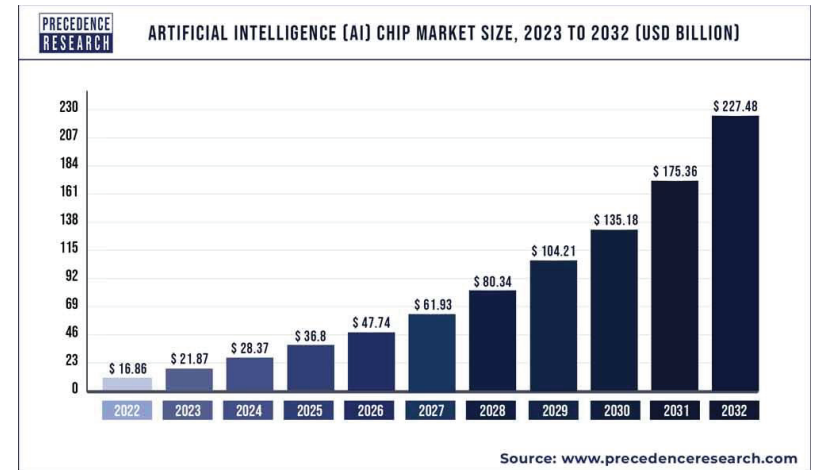
The MPS logo is displayed in a bold, white, sans-serif font against a dark background.

MonolithicPower.com

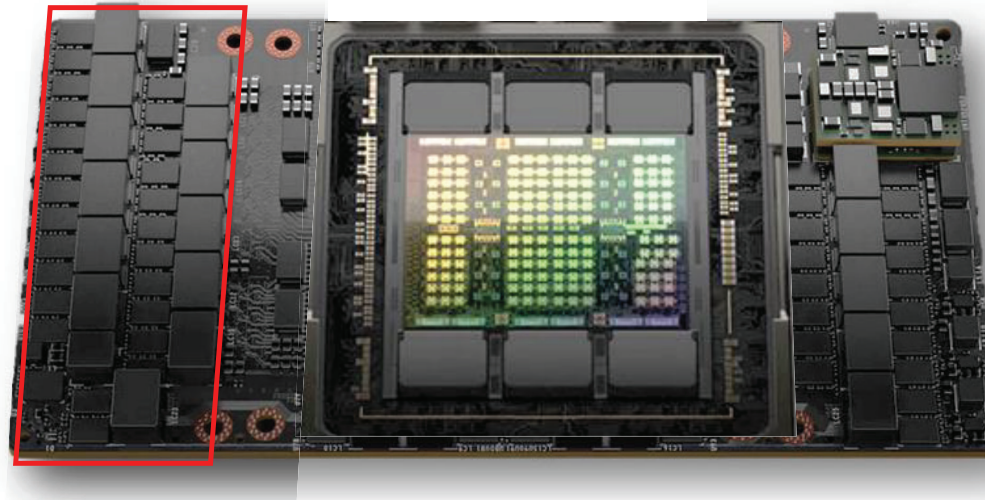
AI Market Trends & Challenges

- AI Market is accelerating
- Power per processor is increasing, seeing upwards >1kW
- Power density is increasing
- Transient performance getting stricter

→Market moving towards Z-axis power delivery (ZPD™)



The Growing Pain



North and South VR removed → 2x PDN
East and West VR push out → additional 2x PDN
4x power delivery loss ↑

MPS in Data Center with AI & HPC

Total Solution : 54V modules & ICs, intelli-modules, multiphase controllers, DrMOS, digital & analog POL, hot-swap, E-fuse and more

→ Reduces design and supplier engagement time

Supply Chain : Inside and outside of China manufacturing, dual sourcing fab & A/T, etc.

→ Lowers continuity of supply risk

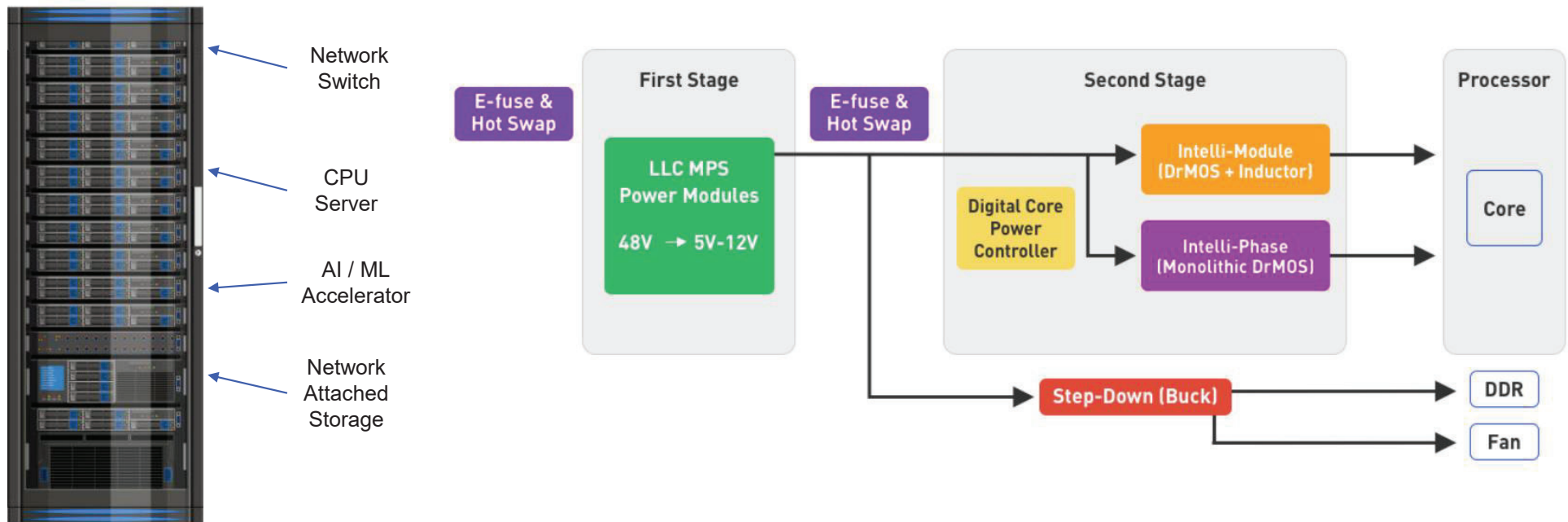
Support : Global AE team drives partnerships with SoC suppliers and system OEMs

→ Closely supporting customer R&D teams to ensure on-time performance of NPI & production schedules

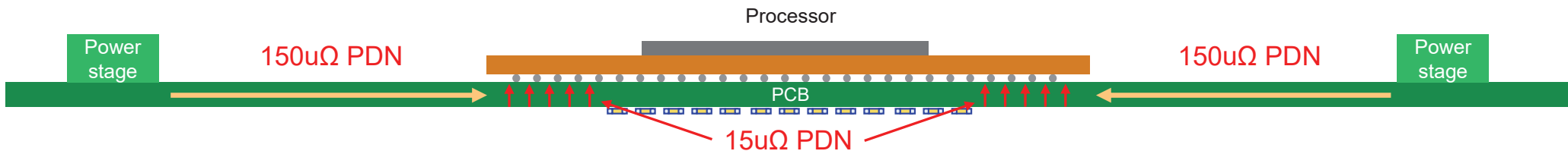


Data center power solutions

<https://www.monolithicpower.cn/cn/products/power-management/48v-data-center.html>



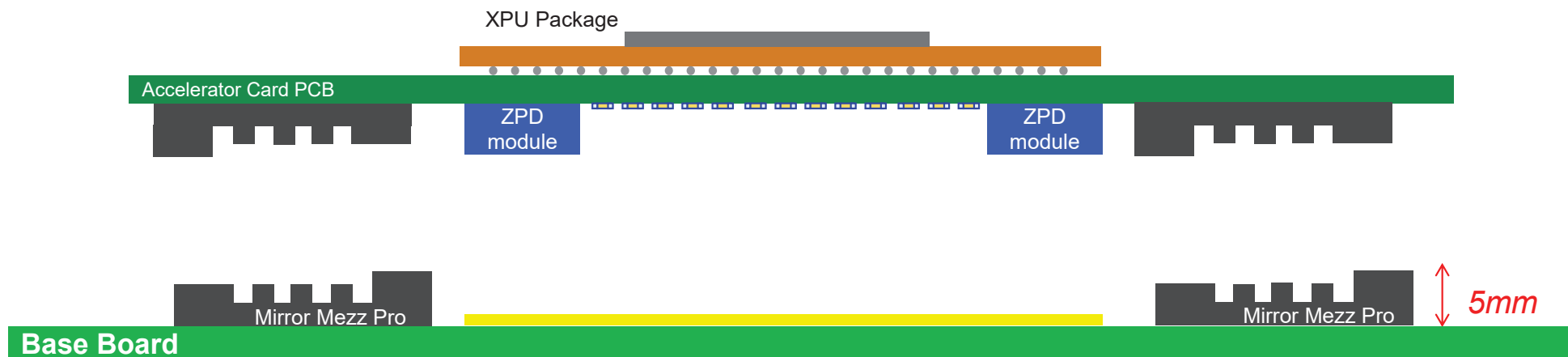
Z-axis Power Delivery Overview



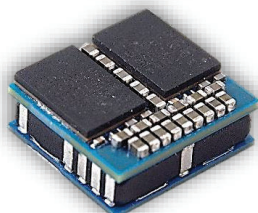
PDN loss is reduced by 10x with ZPD™

- **Performance** *increases* because there is less PDN voltage droop during transient
- **Wattage** loss *decreases* because of less I^2R loss on the PCB
- Therefore, **Perf per Watt** *increases*

ZPD™ Module Height Limited by Mezzanine Connector



5mm Molex Mirror Mezz Pro
is specified in OAM v1.5 for SI.



4mm Height ZPD™ Customized for OAM V1.5

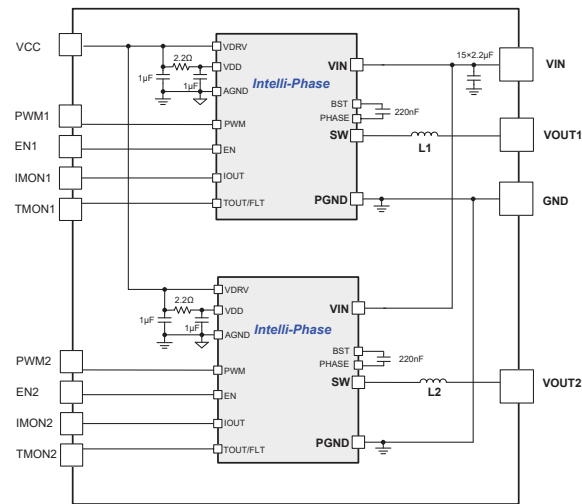
*OCP Accelerator Module (OAM) Design Specification v1.5



Intelli-Module overview

- **High Current Density:** 3D vertical integration structure of components to save layout space by 60%.
- **Excellent Thermal Dissipation:** Strategic placement of active devices to enable top-side cooling.
- **Reduced Design Effort:** Consolidated inductor, driver, and FET modules simplify routing effort for both digital signals and power routing.

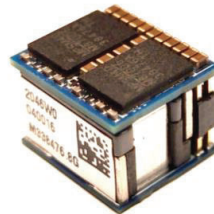
Block Diagram



Features

- ❑ VIN range: 4V to 16V
- ❑ 120-170A Output Current
- ❑ Current & Temperature Sense
- ❑ Current Limit & Over-Temperature Protection (OTP) with Fault Reporting
- ❑ Common Footprint : LGA-72 (9mm x 10mm) package

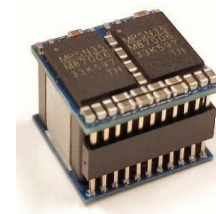
MPC22163



MPC22164



MPC22167



MPC22166 (4mm)



MPC22166 Overview

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package
- Extremely Low Height: Maximum 4mm



Choosing the best architecture

*All 3 cases are CFP or co-layout

- **Mainstream**

- 54V → 13.5V → Processor
- Lateral Power Delivery(LPD) : MPC12106, 4:1 IBC + MPC22167 Intelli-module

- **Best processor transients**

- 54V → 5.4V → Processor
- *Z-Axis Power Delivery (ZPD™)* : MPC1100, 10:1 IBC + MPC22166 Intelli-module

- **Lowest cost**

- Equivalent to mainstream, with down design
- Lateral Power Delivery(LPD) : 1st stage discrete ICs + 2nd stage DrMOS

Thank you

Thank you

Thank you



48V Datacenter Solutions

DC/DC Power Conversion for Datacenter, Open Compute & AI Applications

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Overview

Today's datacenters use an average of 3kW to 5kW per rack to power server, storage, and networking racks. Most are designed to power basic CPUs and to operate at high levels of efficiency. Hence, the traditional 12V power architecture has been widely accepted and implemented (see Figure 1).

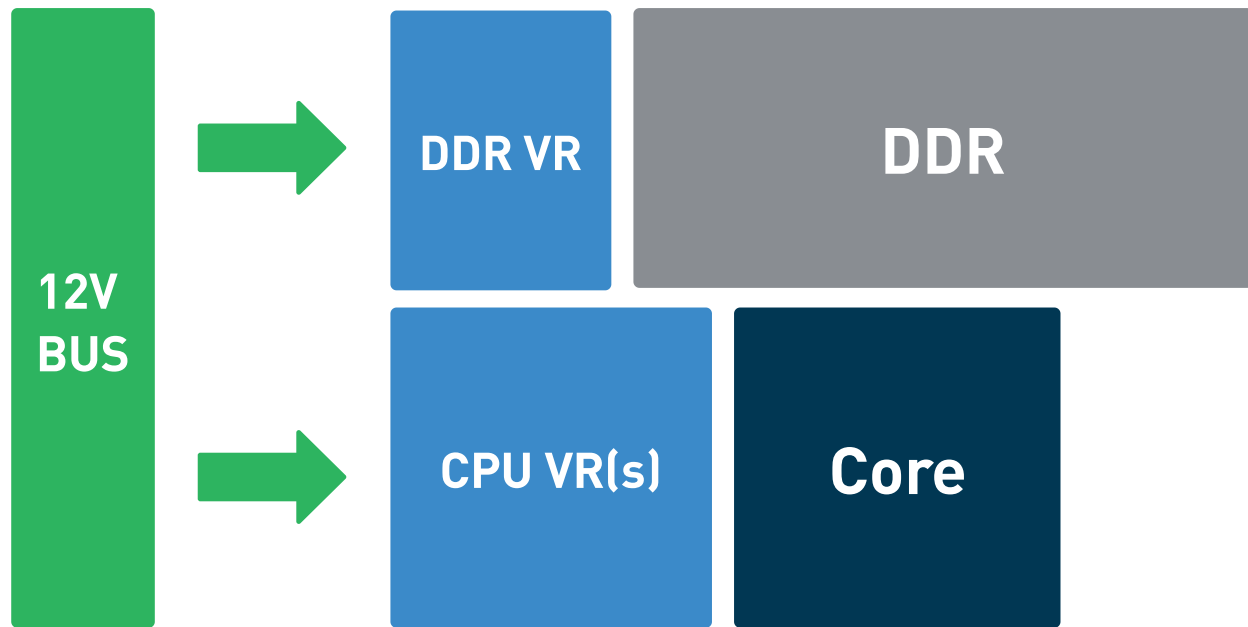


Figure 1: Traditional 12V Power Distribution

As the demand for cloud computing, AI applications, machine learning, high-performance computing, and supercomputers grows, datacenters are evolving to accommodate new, higher power requirements. For example, an early AI market supercomputer required 3200W for the entire power system. The second generation's power demand increased threefold, resulting in 10kW for the entire power system. Since the power lost in distributing high current increases by the square of the current (I^2R), more copper must be used in the backplane or wiring harness to control the distribution losses. Ultimately, this limits the power delivery of the system.

In order to meet the industry's new power requirements, MPS has developed a new power architecture, using a 48V distribution voltage that is capable of a 16x reduction in power distribution losses, in addition to presenting a 48V datacenter solution to address new design challenges.

MPS 48V Power Structure

48V Power Structure: MPS's New 48V to 4.8V System

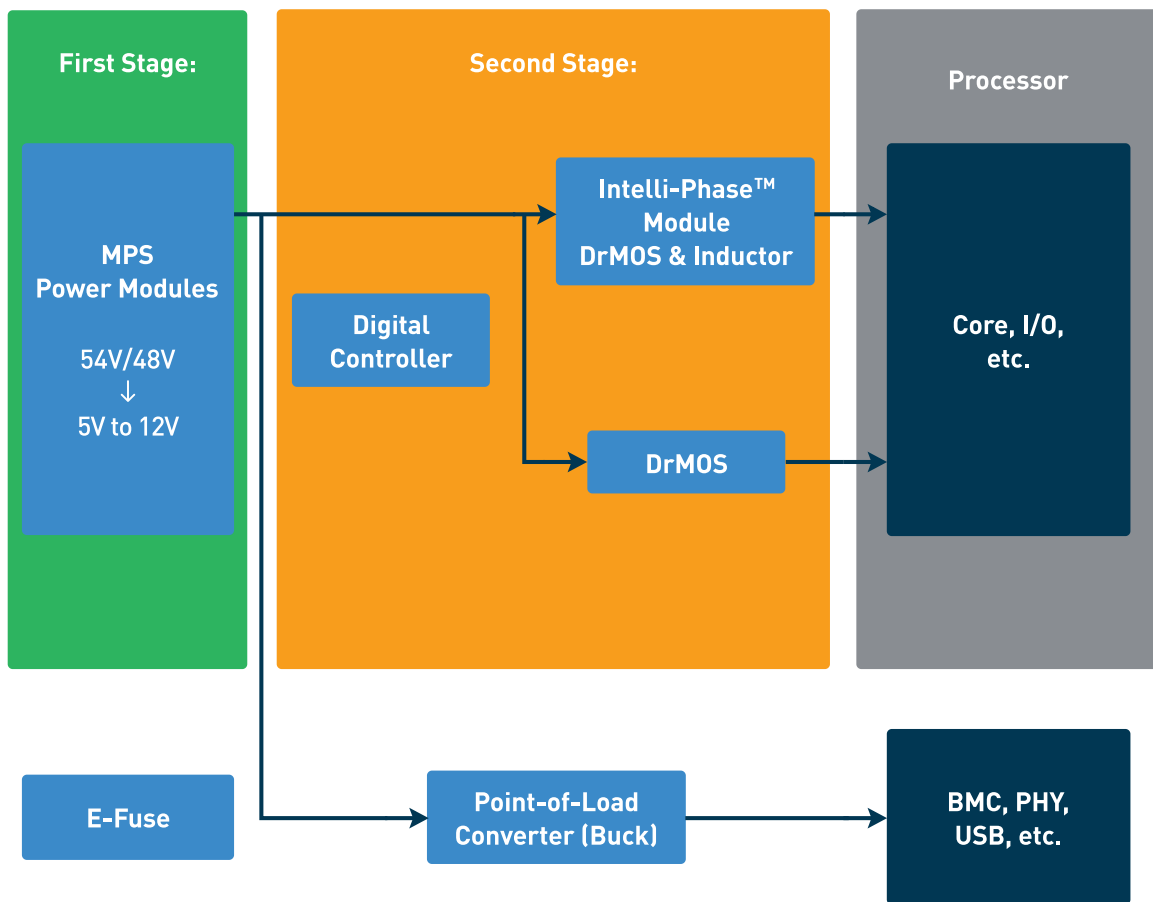


Figure 2: 48V Power Structure – MPS 4V to 6V System

- Total system efficiency is greatly optimized
- First-stage solution: MPS LLC power modules
 - 10:1 input-to-output ratio
 - Offered at various power levels
 - Parallel operation
- Second-stage solution: digital multi-phase controller + Intelli-Phase™ or Intelli-Module
 - The Intelli-Phase™ family offers continuous current ratings up to 90A
 - Intelli-Module solutions offer continuous current ratings up to 170A
- E-Fuse

Intelli-Modules are MPS's latest open-frame power modules, and are part of our second-stage solution. Intelli-Modules can achieve high power density and optimize thermal performance in space-constrained applications.

Intelli-Phase™ is MPS's state-of-the-art power stage technology. MPS's DrMOS uses our monolithic process to drastically improve performance, and incorporates advanced features, including current sense, temperature sense, and fault reporting.

Lateral vs. Z-Axis Power Delivery™

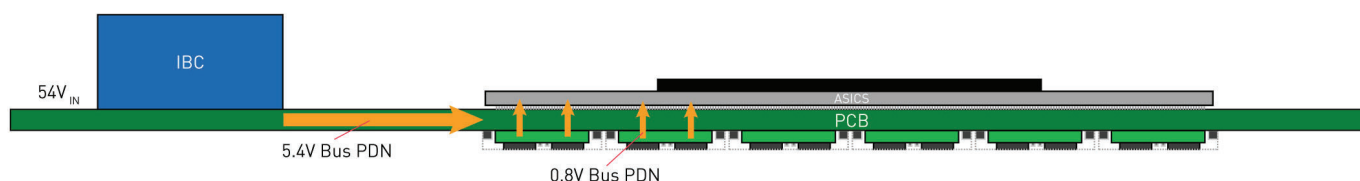
Today's datacenters use a lateral power delivery system, in which the power voltage regulators are placed on the top side of the board around the processor. As the current demand of CPUs and GPUs increases, the distance between the voltage regulator and the point of load become a significant factor for PDN losses. In addition, increased PDN means that the system requires a large amount of output capacitance for the voltage regulators.

Z-Axis Power Delivery™ is a method by which the voltage regulators are placed on the bottom side of the PCB below the processors. This results in a significant reduction (more than 10x) in PDN losses.

Lateral Power Delivery <1000A



Z-Axis Power Delivery™ >1000A



MPS offers power module solutions with <4mm height to fit in the limited space on the bottom side of the PCB.

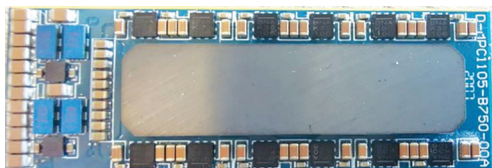
Contact MPS for more information on Z-Axis Power Delivery™ EVBs

First-Stage Solution: Power Modules

MPS offers multiple options for first-stage modules

10:1 IBC Modules

54V to 5.4V

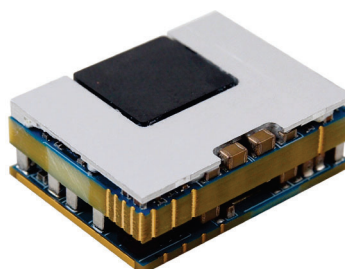


Enables a higher frequency second stage due to lower switching losses, which allows the use of a smaller, shorter inductor

- Best for Z-Axis Power Delivery™ (ZPD)

4:1 IBC Modules

54V to 13.5V



Higher power density, established common footprint in the market

- Best for lateral power delivery (LPD)

All of MPS's IBC modules can be paralleled to reach >1kW power levels.
An application note with additional details is in development.

MPC1100C-54 – High-Efficiency, Non-Isolated, Fixed-Ratio Digital DC/DC 300W Power Module

The MPC1100C-54 is an open-frame digital DC/DC power module with continuous power up to 300W. This module accepts up to a 60V input, and is configured in a 10:1 input-to-output ratio. It offers a digital controller with multiple-time programmability (MTP), which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

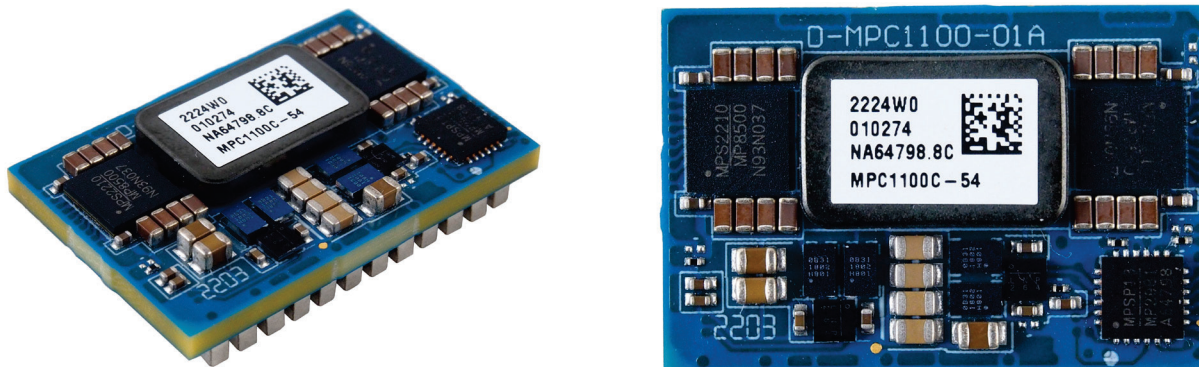


Figure 3: MPC1100C-54

Features:

- Input Voltage Range: 40V to 60V
- 10:1 Input-to-Output Ratio
- Power Level: Up to 300W
- Power Density: 1700W/in³
- LLC Topology
- Compatible with Serial Interface
- Supports Parallel Operation
- Built-In MTP to Store Custom Configurations
 - Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided
- Great for High-Current and High-Power Applications
- Pin-to-Pin Compatible with the MPC1102-54

Dimensions:

27mmx18mmx6.1mm

Part Number	V_{IN} (V)	V_{OUT} Range (V)	Power Density (W/in ³)	P_{OUT} (W)	Peak Efficiency (%)	Efficiency @ 300W (%)
MPC1100C-54-0002	40 to 60	4 to 6	1700	300	96.7	95.3

MPC11057-54-0750 – High-Efficiency, Non-Isolated, Fixed-Ratio Digital DC/DC 750W Power Module

The MPC11057-54-0750 is an open-frame digital DC/DC power module with continuous power up to 750W (see Figure 4). This module accepts up to a 60V input, and is configured in a 10:1 input-to-output ratio. It offers a digital controller with multiple-time programmability (MTP), which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

Features:

- Input Voltage Range: 40V to 60V
- 10:1 Input-to-Output Ratio
- Power Level: Up to 750W
- Power Density: 1840W/in³
- LLC Topology
- Compatible with Serial Interface
- Built-In MTP to Store Custom Configurations
 - V_{IN} Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided
- Great for High-Current and High-Power Applications

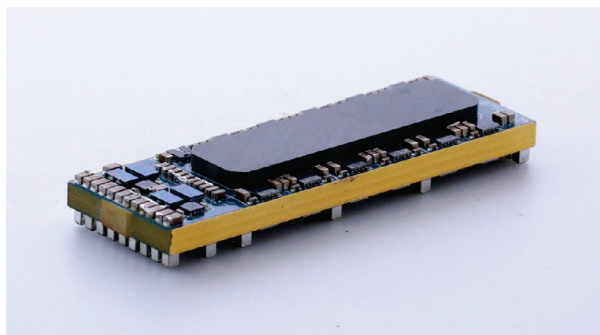


Figure 4: MPC11057-54-0750-xxxx

Dimensions:

53mmx18mmx7mm

Part Number	V_{IN} (V)	V_{OUT} Range (V)	Power Density (W/in ³)	P_{OUT} (W)	Peak Efficiency (%)	Efficiency @ 750W (%)
MPC11057-54-0750-xxxx	40 to 60	4 to 6	1840	750	97.6	96.9

MPC12106-54-0750 – High-Efficiency, Non-Isolated, Fixed-Ratio Digital DC/DC 750W Power Module

The MPC12106-54-0750 is an open-frame digital DC/DC power module with continuous power, up to 750W. This module accepts up to a 60V input, and is configured in a 4:1 input-to-output ratio. It offers a digital controller with multiple-time programmability (MTP), which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

Features:

- Input Voltage Range: 40V to 60V
- 4:1 Input-to-Output Ratio
- Power Level: Up to 750W
- Power Density: 3310W/in³
- LLC Topology
- Compatible with Serial Interface
- Built-In MTP to Store Custom Configurations
 - V_{IN} Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided
- Great for High-Current and High-Power Applications

Dimensions:

24mmx18mmx8.6mm

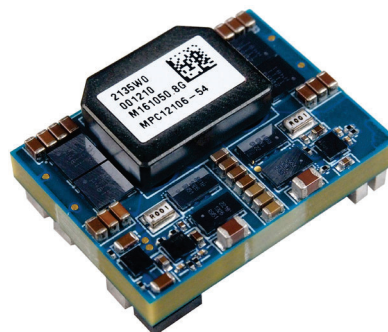


Figure 5: MPC12106-54-0750

Part Number	V _{IN} (V)	V _{OUT} Range (V)	Power Density (W/in ³)	P _{OUT} (W)	Peak Efficiency (%)	Efficiency @ 750W (%)
MPC12106-54-0750-xxxx	40 to 60	10 to 15	3310	750	98.1	97.5

Second-Stage Solution: Digital Multi-Phase Controller + Intelli-Phase™ or Intelli-Module

MPS has a robust product portfolio of digital multi-phase controllers and Intelli-Phase™ and Intelli-Module solutions. For more information, visit monolithicpower.com/en/products/power-management/data-center.html

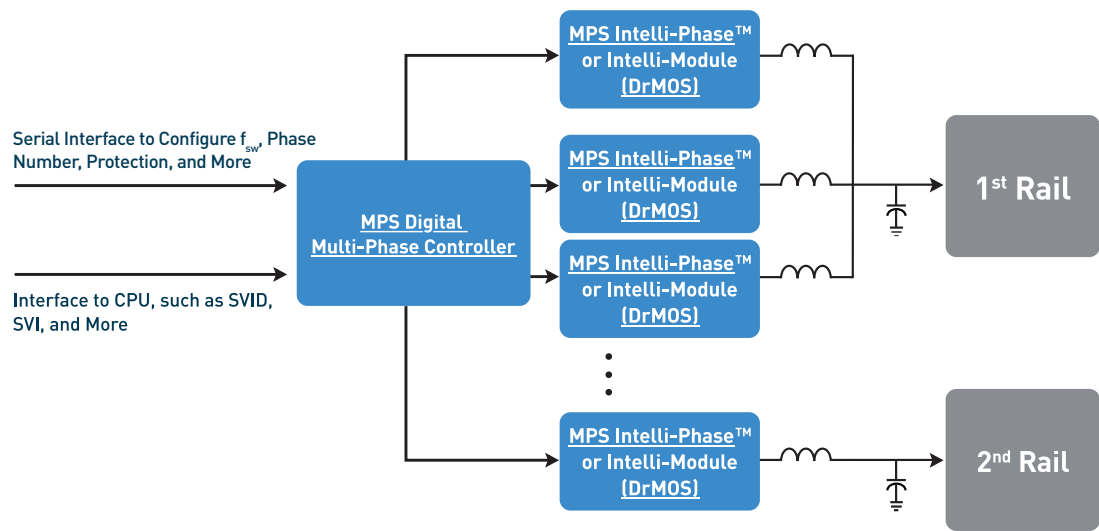


Figure 7: Second-Stage Digital Multi-Phase Controller + Intelli-Phase™ or Intelli-Module Block Diagram

Digital Multi-Phase Controller Features:	Intelli-Phase™ Features:	Intelli-Module Features:
- Patented Constant-On-Time (COT) Control - Multiple Loops, Flexible Phase Assignment - Interface Options: SVID, SVI2/3, OVR, or Serial Interface - Fault Handling and Telemetry Capabilities	- Family of Devices Ranges from 10A to 90A - Wide f_{sw} Range: Balance between Transient and Efficiency - Built-In ZCD, OCP, NOCP, OTP, and SCP - Minimal External Components	- High Power Density in Space-Constrained Applications - Flexible Output Assignment - Support Parallel Operation for Higher Power Delivery - Optimized for Top-Side Cooling

Power Structure	Type of Product	Part Number	Part Description
12V & MPS's new 4V to 6V architecture	Digital multi-phase controller	MP2888A	Single-loop, 10-phase digital multi-phase controller
		MP2882 (contact MPS for samples)	Dual-loop, 16-phase digital multi-phase controller
		MP2972 (contact MPS for samples)	Dual-loop, 12-phase digital multi-phase controller
12V architecture	DrMOS Intelli-Module	MP87006	16V, 90A Intelli-Phase™ in 4mmx6mm package
		MP87000	16V, 90A Intelli-Phase™ in 5mmx6mm package
		MPC22163 (contact MPS for samples)	16V, 130A Intelli-Module in 9mmx10mmx8mm footprint
		MPC22164 (contact MPS for samples)	16V, 130A Intelli-Module in 9mmx10mmx4mm footprint
MPS's new 4V to 6V architecture	DrMOS	MP86972 (contact MPS for samples)	12V, 60A Intelli-Phase™ in 3mmx6mm package
		MP86976 (contact MPS for samples)	7V, 70A Intelli-Phase™ in 3mmx6mm package

MPC22163-130 – Two-Phase Intelli-Module with Quiet Switcher™ Technology

The MPC22163 is a non-isolated, step-down power module with 130A of continuous peak output current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW+ of power. The 8mm maximum height makes it suitable for many applications, such as OAM form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Top-Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

Dimensions:

9mmx10mmx7.5mm

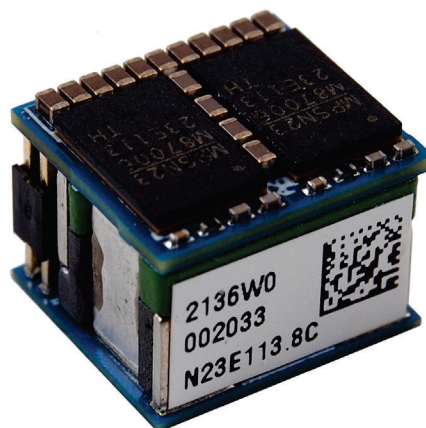


Figure 8: MP22163-130

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22163-130	4 to 16	0.5 to 2	130	600 to 1500	2475	88.7

MPC22166-130 – Two-Phase Intelli-Module with Quiet Switcher™ Technology

The MPC22166 is a non-isolated, step-down power module with 130A of continuous peak output current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW+ of power. The 4mm maximum height makes it suitable for many applications, such as Z-Axis Power Delivery™ and PCIe form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Top-Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package
- Extremely Low Height: Maximum 4mm
- Pin-to-Pin Compatible with the MPC22163

Dimensions:

9mmx9.9mmx3.9mm

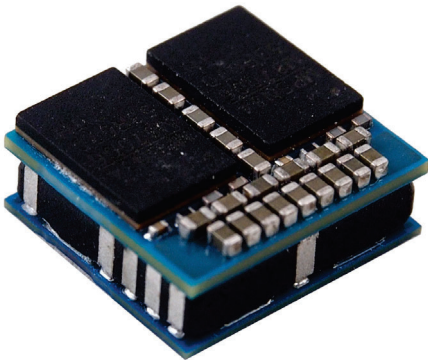


Figure 9: MP22166-130

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in³)	Peak Efficiency (%)
MPC22166-130	4 to 16	0.5 to 2	130	600 to 1200	5034	90

MPC22167 – 130A, Two-Phase, Intelli-Module™ with Quiet Switcher™ Technology

The MPC22167 is a non-isolated, step-down power module with 130A of continuous peak output current. This module integrates driver MOSFETs and an inductor in a compact package to save board space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 8mm maximum height makes it suitable for a wide range of applications, such as OAM form factor boards.

Features:

- Input Voltage Range: 10V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Top-Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

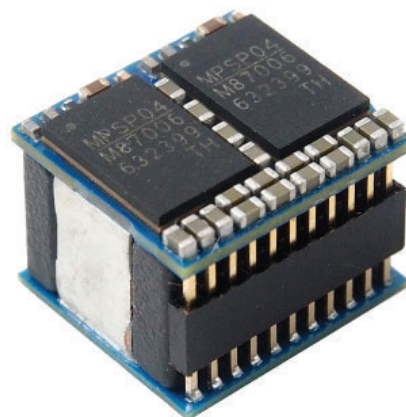


Figure 10: MPC22167

Dimensions:

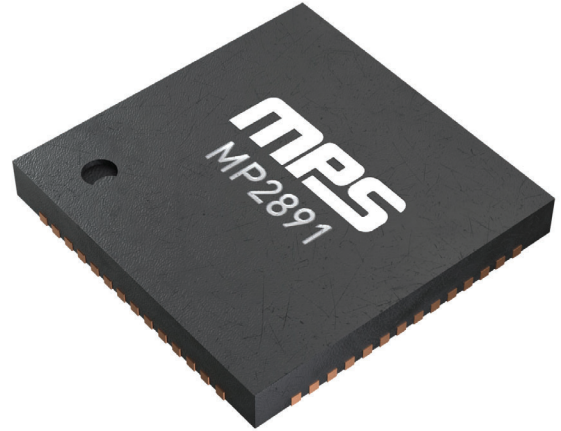
9mmx9.9mmx7.65mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22167-130	4 to 16	0.5 to 2	130	500 to 1200	2475	91.3

MP2891 – Dual-Loop, OVR4i+ 16-Phase Controller

Features:

- Highest Phase Count
 - Dual-Loop, 16-Phase Controller Provides Power for NVIDIA PWM-VID Core Power
 - Max 16 Phases for Rail 1 and 8 Phases for Rail 2
- Digital Control
 - Digital Loop Compensation and Load-Line Regulation
 - Nonlinear Control Reduces Output Capacitor Requirements for Transients
- Digital Configurability and Monitoring
 - Compatible with Serial Interface
 - V_{OUT} , I_{OUT} , V_{IN} , P_{IN} , and Temperature Telemetry
 - Intelli-Phase™ Fault Diagnosis
 - Integrated NVM to Store VR Configurations and for Black Box Capability
- Design Flexibility
 - Switching Frequency Up to 3MHz
 - Automatic Phase-Shedding Improves Light-Load Efficiency
 - Configurable Voltage, Current, and Temperature Protection



Package:
QFN-56 (7mmx7mm)

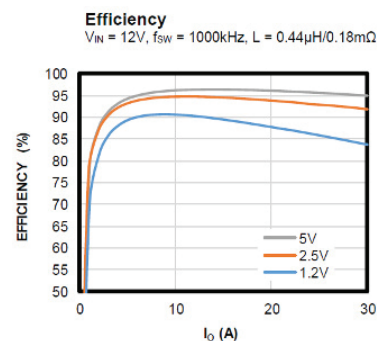
MPQ8645P – 16V, 30A, Scalable, Digital, Synchronous Step-Down Converter with Serial Interface

Features:

- Digital Point-of-Load (POL)
 - V_{IN} , V_{OUT} , I_{OUT} , and Temperature Telemetry
 - Rich Feature Selection
 - User-Accessible NVM to Save Configurations
- Flexible for Multiple Rails in System
 - Scalable Up to 8 Phases
 - Input Voltage Range: 3.1V to 16V
 - Supports 3.3V, 5V, and 12V Inputs
 - Supports Zero-ESR, MLCC Output Capacitors
 - Configurable Current Limit Offers Flexible Inductor Choice
- Adaptive Constant-On-Time Control
 - Minimizes Number of Required Output Capacitors
 - 0.5% Accurate Reference



Package:
TQFN-25 (4mmx5mm)



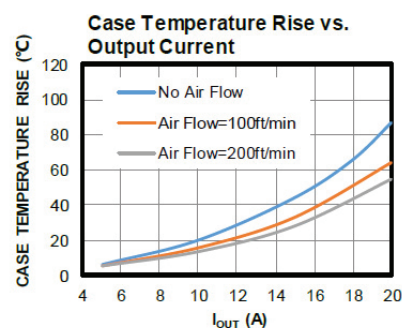
MP5048 – 60V, 15A, 7m Ω $R_{DS(ON)}$ Hot-Swap Intelli-Fuse Solution

Features:

- 60V E-Fuse with Highest Industry Current
- Flexible for Multiple Phases in System
 - Input Voltage Range: 24V to 60V
 - Scalable for Standalone or Parallel Operation
 - Configurable Output Voltage Soft Start Supports Inrush Current Control
 - 1.5% IMON Reporting Accuracy Supports Accurate System Power Reporting
- Full System Protections
 - Configurable Over-Current Protection (OCP) Limit
 - Accurate On-Die Temperature Sensing and Thermal Shutdown
 - Pin-Configurable Latch-Off and Hiccup Protections
 - Fault Pin Indicator (FLTBI)



Package:
QFN-30 (5mmx5mm)



Second-Stage Module vs. Traditional Discrete Solution

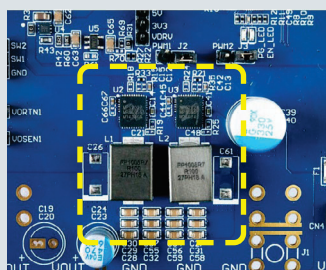
As the current demand of CPUs and GPUs increases and package sizes expand, designers are often left with very small areas on the board to fit multiple phases of voltage regulators (VRs). The Intelli-Module approach to VR design increases power density by 2.5x and allows placement of the voltage regulators closer to the processor to reduce PDN losses.

Intelli-Modules: DrMOS + Inductor



Intelli-Module Solution Offers the Best Combo of Efficiency and Size
90mm² Layout Area

DrMOS (Discrete Approach)



Traditional VR Solution
225mm² Layout Area

OAM Form Factor Power Solutions

MPS provides end-to-end power solutions following OCP standards for common designs, such as OAM form factors. These capabilities are demonstrated with this reference design (max 2000A), which features our 4:1 first-stage modules and second-stage Intelli-Modules.

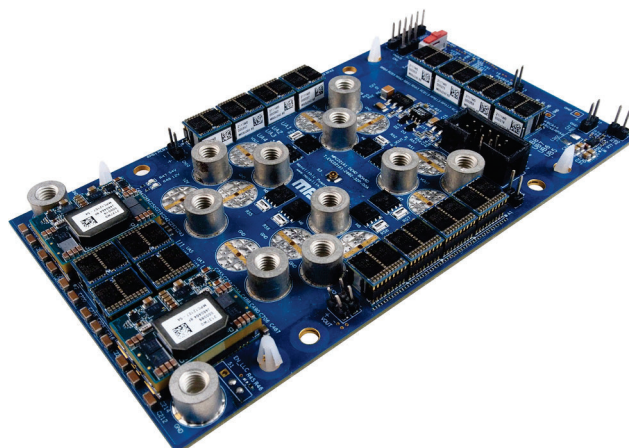
2000A Reference Design

Specs:

- 54V Input Voltage, 0.8V Nominal Output Voltage
- 1000A Output Current, Max 2000A
- Embedded Load Slammer for Transient Testing
- Second Stage Uses the MP2891 16-Phase Controller and the MPC22167, the Next Generation of the MPC22163 Module with the Same Footprint and Higher Efficiency
- This 2000A EVB Is Now Available for Sampling

AI Accelerator Card:

- OAM
- SXM
- etc.

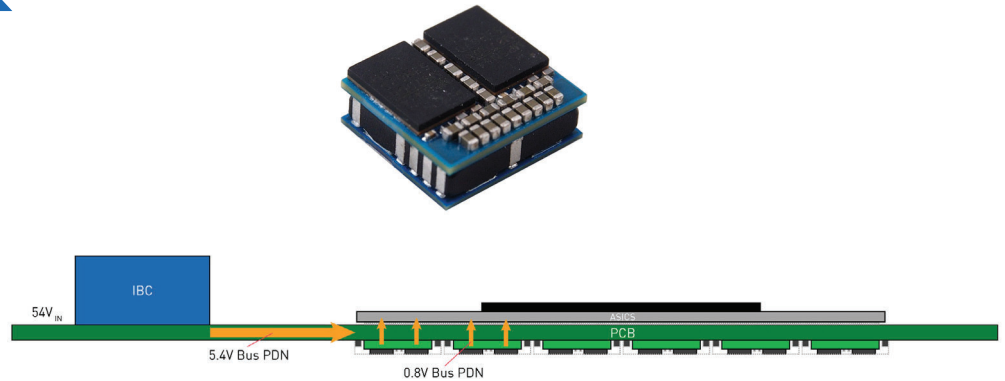


For samples and more information on Z-Axis Power Delivery™ solutions, contact MPS

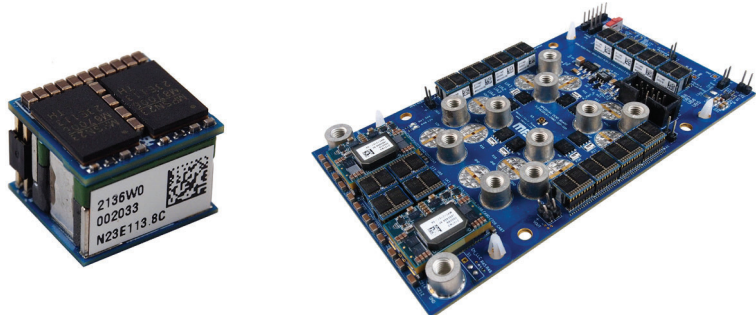
Adapting Power Solutions to the Increased Current Requirement of AI Processors

Increased Current Requirements

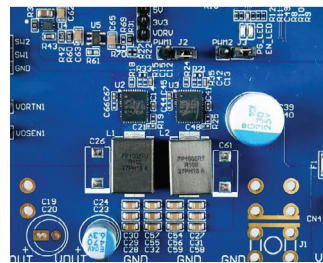
Z-Axis Power Delivery™
Ex: MPC22166



Lateral Power Delivery Using Modules
Ex: MPC22163



Lateral Power Delivery Using DrMOS
Ex: MP87006



Quality Assurance & Reliability Commitment

The MPS Quality Assurance organization develops, coordinates, and champions strategic quality initiatives throughout MPS Inc., its foundries, and subcontractors. Its mission is to enable MPS to design, develop, manufacture, and deliver products to our customers with world-class quality and reliability that meet and exceed our customers' expectations.

MPS and Its Supplier Quality Systems and Certificates:

- ISO9001:2008 (MPS)
- EU RoHS/HF/REACH Compliant (MPS)
- Sony Green Partner (MPS & Suppliers)
- TS16949 (Suppliers)
- ISO14001 (Suppliers)

Product Quality:

- Automotive Products Qualified per AEC-Q100 Standard
- Standard Products Qualified per JEDEC and Military Standard
- Reliability Failure Rate <10FIT
- Product Quality Level <1.0ppm

Quality Control and Monitor:

- On-Site Foundry and Assembly Teams for Real-Time Actions
- Quarterly Supplier Quality Review and Annual Supplier Audit
- Short-Term Reliability Monitor Test – Daily
- Long-Term Reliability Monitor Test – Monthly
- Real-Time Engineering Actions on Monitor Failure
- Quarterly Reliability Monitor Reports

About MONOLITHIC POWER SYSTEMS

Who we are

We are creative thinkers. We break boundaries. We take technology to new levels. As a leading international semiconductor company, Monolithic Power Systems (MPS) creates cutting-edge solutions to improve the quality of life with green, easy-to-use products.

What we do

We make power design fun! With our innovative proprietary technology processes, we thrive on reimagining and redefining the possibilities of high-performance power solutions in industrial applications, telecom infrastructures, cloud computing, automotive, and consumer applications.

Where we come from

It started with a vision. Michael Hsing, pioneering engineer and CEO, founded Monolithic Power Systems, Inc. in 1997 with the belief that an entire power system could be integrated onto a single chip. Under his leadership, MPS has succeeded not only in developing a monolithic power module that truly integrates an entire power system in a single package, but also it continues to defy industry expectations with its patented groundbreaking technologies.

Our values

We cultivate creativity

As a company we believe in creating an environment that encourages and challenges our employees to collaborate and think outside the box to excel beyond their preconceived capabilities.

We do not accept the status quo

We do not believe in limitations. It is not about what is, but what can be. Possibilities are endless at MPS.

We are passionate about sustainability

It's about the future. From materials to finances, we are committed to conservation. We will not tolerate waste in an effort to improve and preserve the quality of life.

We are committed to providing innovative products to our customers

Let us do the heavy lifting. We relentlessly strive to make system design versatile and effortless to meet our customers' specific needs. We'll do the work, so our customers can have the fun!

CONTACT & ORDERING



**FAST
RESPONSE**

Quotes, Availability, Engineering Support & Samples
MonolithicPower.com/Quote-Samples-Support

US Offices

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5808 Lake Washington Blvd. NE
Kirkland, WA 98033, USA
Tel: +1 425-296-9956
- **MPS San Jose**
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Tel: +1 408-826-0600
- **MPS Detroit**
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Livonia, MI 48152, USA
Tel: +1 248-907-0222
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2065 S. Cooper Road, Suite 3
Chandler, AZ 85826, USA

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08029 Barcelona, Spain
Tel: +34-931-815-400
- **MPS Ettenheim**
Gutenbergstrasse 4,
77955 Ettenheim,
Germany
- **MPS Munich**
Alte Landstrasse 25
85521 Ottobrunn, Germany
Tel: +49-89-80913512-0
- **MPS Lausanne**
Route de Lully 5 A
1131 Tolochenaz, Switzerland
Tel: +41-21-805-0100

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Online Order Support
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Processing Zone West Hi-Tech Zone
Chengdu, Sichuan, 611731
Tel: +86-28-8730-3000
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- **MPS China Shenzhen**
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Square Branch North,
Binhe Avenue South, Futian District
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Tokyo 163-0231, Japan
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- **MPS Japan Osaka**
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#03-38, Singapore 529482
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- **MPS Korea**
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Sampyeong-dong 624 5pilgi, Bundang-gu,
SeongNam-city, Kyunggi-do, Korea
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- **MPS India**
Unit G-12, Prestige Towers,
No 99 / 100, Residency Road,
Bangalore 560025
Tel: +91-80-4124-0312 / 20
- **MPS Investor Relations**
monolithicpower.com/investors



MPS 48V Datacenter Solutions

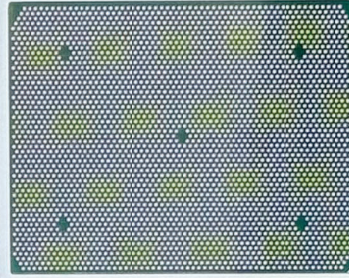
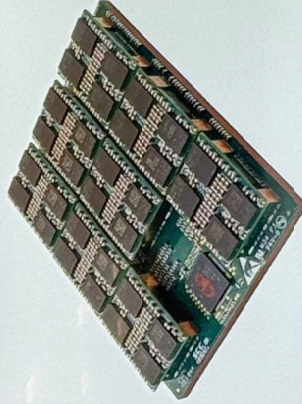
DC/DC Power Conversion for Datacenter,
Open Compute & AI Applications

Z-Axis Power Delivery

Custom ZPD™ Module

MPC2M110-L:

- Fully Customized to Optimize System Performance
- Integration of All Components: Controller, DrMOS, Inductors, and Capacitors
- Direct Power Ball Mapping to Minimize PDN
- Highest Power Density





Search

MPC22166-D-130

5mm, 130A, Dual-Phase Intelli-Module™ with Quiet Switcher™ Technology

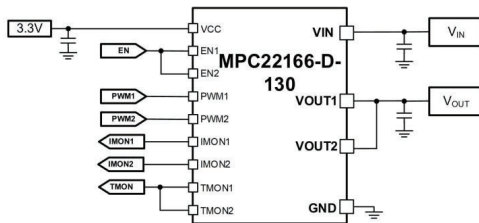
Pre-Release

[Overview](#) [Resources](#) [Design Resources](#) [Technical Forum](#) [Quality](#)

MPC22166-D-130

5mm, 130A, Dual-Phase Intelli-Module™ with Quiet Switcher™ Technology

Pre-Release



Description

The MPC22166-D-130 is a dual-phase, non-isolated, high-efficiency, step-down power module. It integrates driver MOSFETs (DrMOS) and inductors to provide high efficiency in a small footprint, which is ideal for AI and server applications.

The MPC22166-D-130 provides two outputs that can support dual-rail operation and single-rail operation. It can achieve up to 130A (65A/phase) of output current (I_{OUT}) across a wide input voltage (V_{IN}) range.

The MPC22166-D-130 offers many features to simplify system design. The device works with tri-state pulse-width modulation (PWM) signal controllers, and provides Accu-Sense™ current sensing and temperature sensing to monitor the inductor current (I_L) and report the junction temperature (T_J), respectively.

The MPC22166-D-130 is available in an LGA-72 (9mmx9.9mmx4.75mm) package.

Show less ^

Features & Benefits

- Quiet Switcher™ Technology (QST) Limits Peak Switching Voltages
- Wide 4V to 16V Input Voltage (V_{IN}) Range
- Up to 130A Output Current (I_{OUT})
- Accu-Sense™ Current Sense (CS)
- Temperature Sense
- Compatible with Tri-State Pulse-Width Modulation (PWM) Signals
- Over-Current Protection (OCP)
- Over-Temperature Protection (OTP)
- Fault Reporting
- Typical Weight: 2.1g
- Available in an LGA-72 (9mmx9.9mmx4.75mm) Package

Active Part Numbers: MPC22166-D-130-T

MPC22166-D-130 Resources

Page 1 of 7

NETWORK SWITCH

APPLICATION

[Home > Applications > Datacenter > Network Switch](#)
[Simplify management, reduc...](#)

POWER SOLUTION FOR SWITCHES

APPLICATION

PROCESSOR CORE POWER INTELLI-MODULE (DRMOS...

PRODUCT CATEGORY

INTELLI-

PRODU

Need more support?

[Get Technical Support or Quote](#)

Design Resources

Symbol, Footprint & 3D Model

30+ more formats

 Symbols (36)

 Footprint (34)

 3D Models (15)

EDA model is not yet available for this part.
Please enter your email address and we will notify you when it is released.

Submit Request

Technical Forum

Questions answered by MPS engineers in 24h

MPC22166-B-130 step down power module

what is the difference between MPC22166-B-130 and MPC22166-D-130, are they both same.

LIGHTING SOLUTIONS

🕒 Latest activity 5 months ago

💬 3 Comments

[Get technical support](#)

Quality Policy, Certificates & Reliability Reports

All MPS parts are lead-free, halogen-free and adhere to the RoHS directive. See Environmental section for details.

[SEE ALL QUALITY DOCUMENTS](#) >



Quality
Policy

[DOWNLOAD](#)



ISO
Certificate

[DOWNLOAD](#)



RoHS/Lead
Free Policy

[DOWNLOAD](#)



MPC22163-130

130A, Dual-Phase Intelli-Module™ with Quiet Switcher™ Technology

DESCRIPTION

The MPC22163-130 is a dual-phase, non-isolated, high-efficiency step-down power module. It integrates driver MOSFETs (DrMOS) and inductors to provide high efficiency and a small footprint, which is ideal for server applications.

The MPC22163-130 has two independent outputs, which can be used separately or in parallel. It can achieve up to 130A (65A/phase) of output current (I_{OUTx}) across a wide input voltage (V_{IN}) range.

The MPC22163-130 offers many features to simplify system design. The device works with tri-state pulse-width modulation (PWM) signal controllers, has Accu-Sense™ current sensing to monitor the inductor current (I_L), and has temperature sensing to report the junction temperature (T_J).

The MPC22163-130 is available in an LGA-72 (9mmx10mmx7.65mm) package.

FEATURES

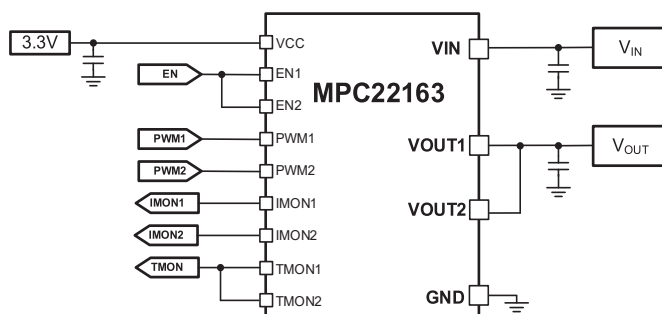
- Quiet Switcher™ Technology (QST) Limits Peak Switching Voltages
- Wide 4V to 16V Input Voltage (V_{IN}) Range
- Up to 130A Continuous Output Current (I_{OUTx})
- Accu-Sense™ Current Sense (CS)
- Temperature Sense
- Compatible with Tri-State Pulse-Width Modulation (PWM) Signals
- Over-Current Protection (OCP)
- Over-Temperature Protection (OTP)
- Fault Reporting
- Typical Weight: 2.6g
- Available in an LGA-72 (9mmx10mmx7.65mm) Package

APPLICATIONS

- Servers and Computing
- Field-Programmable Gate Arrays (FPGAs) and ASIC Core Power
- Graphic Card Core Regulators

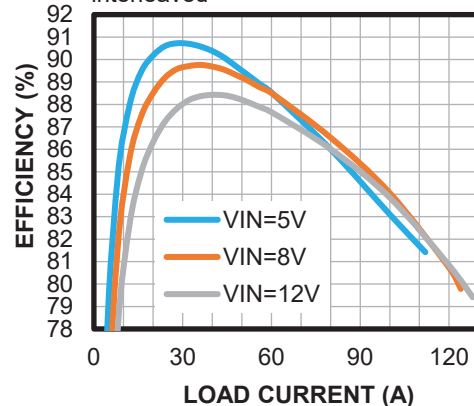
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TYPICAL APPLICATION



Efficiency vs. Load Current

$V_{OUT} = 0.8V$, $f_{SW} = 700kHz$,
interleaved



ORDERING INFORMATION

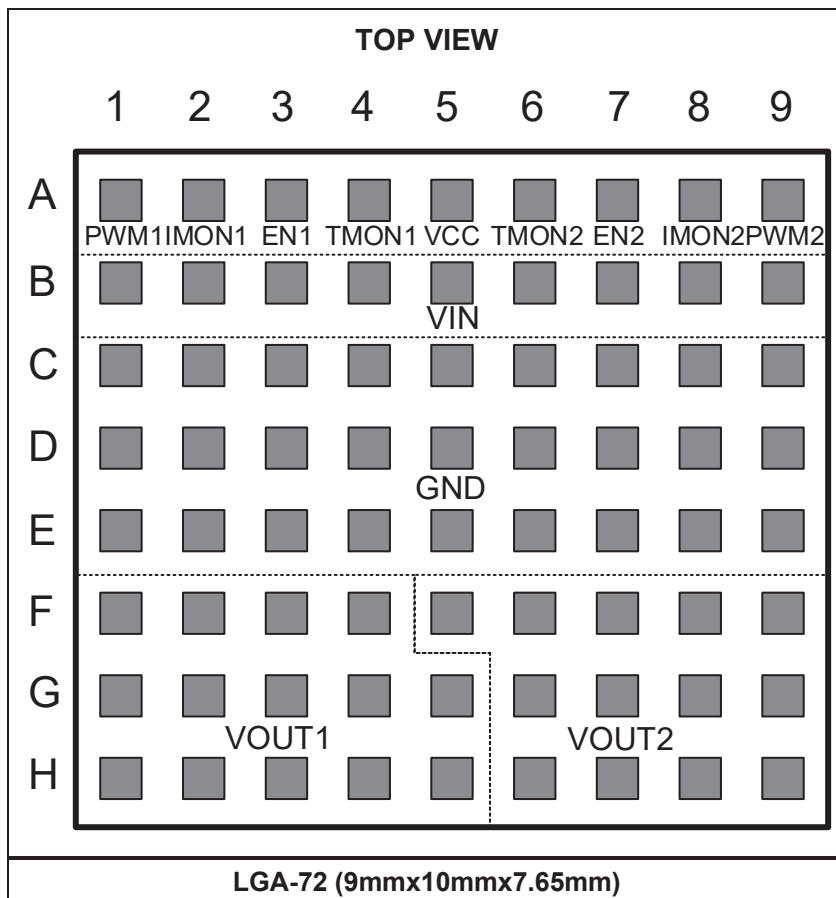
Part Number	Package	Side Label	MSL Rating
MPC22163-130	LGA-72 (9mmx10mmx7.65mm)	See Below	3

* For Tray, add suffix -T (e.g. MPC22163-130-T).

SIDE LABEL

Date Code
Vendor's Serial Number
LOT ID

PACKAGE REFERENCE



Notice: The information in this document is subject to change without notice. Please contact MPS for current specifications. Users should warrant and guarantee that third-party Intellectual Property rights are not infringed upon when integrating MPS products into any application. MPS will not assume any legal responsibility for any said applications.

48V DATACENTER SOLUTIONS

DC/DC Power Conversion for Datacenter,
Open Compute & AI Applications

MPS
MonolithicPower.com

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Overview

Today's datacenters use an average of 25kW per rack to power AI accelerators, servers, storage, and networking equipment. Most are leveraging a legacy architecture of powering basic CPUs with high levels of efficiency. Hence, the traditional 12V power architecture has been widely accepted and implemented (see Figure 1).

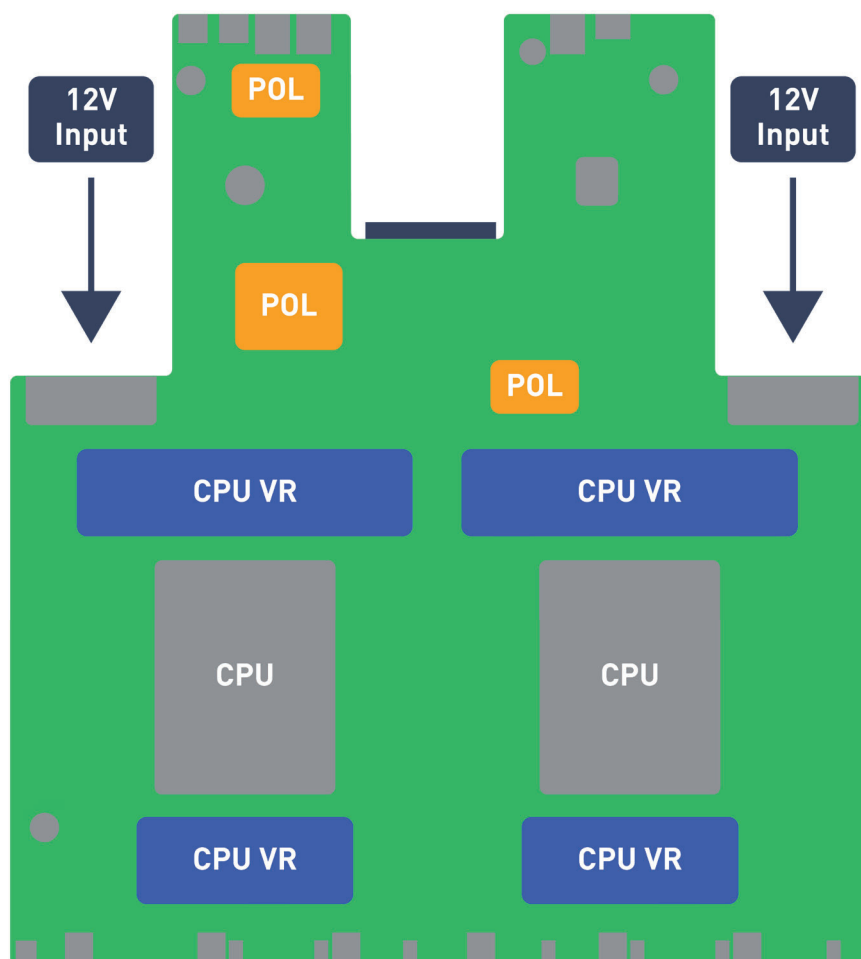


Figure 1: Traditional 12V Power Distribution

To support machine learning and high-performance computing workloads, there is an increasing shift towards a higher percentage of AI accelerator content in new datacenter builds. As a result, datacenters' power architecture is evolving to accommodate new, higher power requirements. For example, the next-generation rack populated with multiple AI accelerator trays will consume over 100kW of power. At this power level, the distribution losses in a 12V system will increase by the square of the current (I^2R), and more copper must be used in the backplane or wiring harness to control the distribution losses. Ultimately, this limits the power delivery capability of the system. In order to meet the industry's new power requirements, MPS has developed solutions for a more efficient power architecture, using a 48V distribution voltage that is capable of a 16x reduction in power distribution losses.

MPS 48V Power Structure

Two-Stage, 48V Processor Architecture

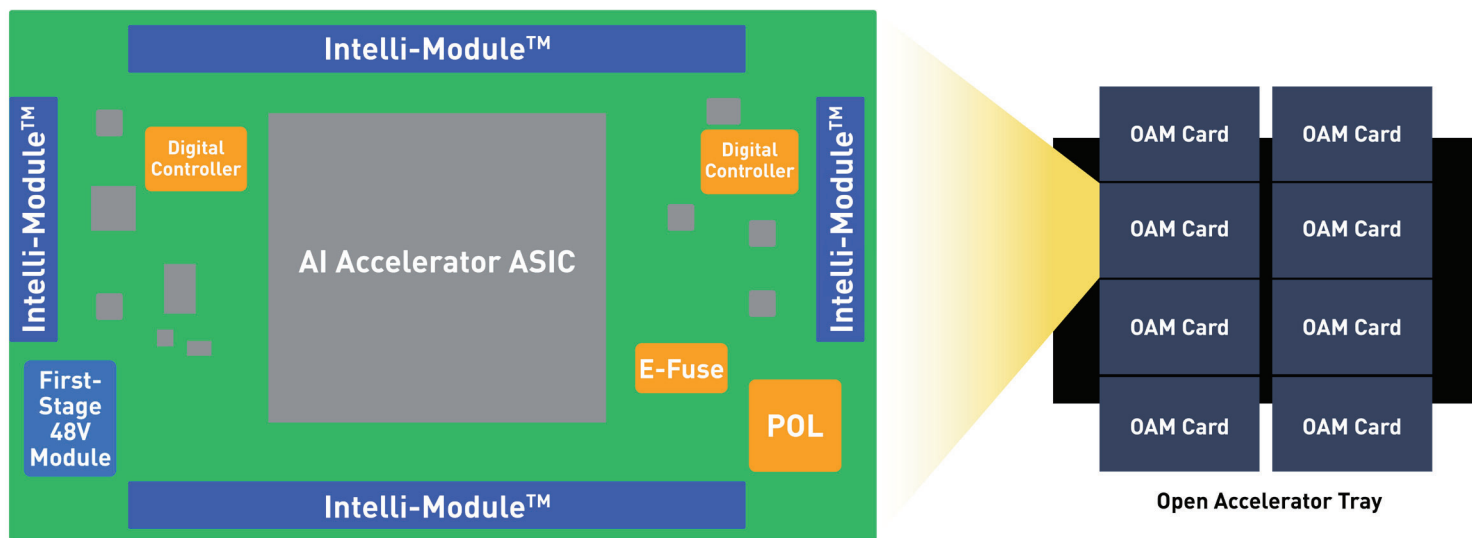


Figure 2: OAM Card with 48V Input

- Total system efficiency is optimized
- First-stage solution: MPS power modules
 - Available with regulated output and fixed input-to-output ratios
 - Offered at various power levels
 - Support for parallel operation
- E-Fuse – Provides system protection from excessive voltage or current point-of-load (POL) and generates system voltages to power interface circuits
- Second-stage solution: digital multi-phase controller + Intelli-Phase™ or Intelli-Module™
 - The Intelli-Phase™ family offers continuous current ratings up to 90A
 - Intelli-Module™ solutions offer continuous current ratings up to 170A

Intelli-Phase™ is MPS's state-of-the-art power stage technology. These DrMOS devices use our monolithic process to drastically improve performance, and incorporate advanced features, including current sense, temperature sense, and fault reporting.

Intelli-Modules™ are MPS's latest open-frame power modules, and are part of our second-stage solution. Intelli-Modules™ can achieve high power density and optimize thermal performance in space-constrained applications.

Lateral vs. Z-Axis Power Delivery™

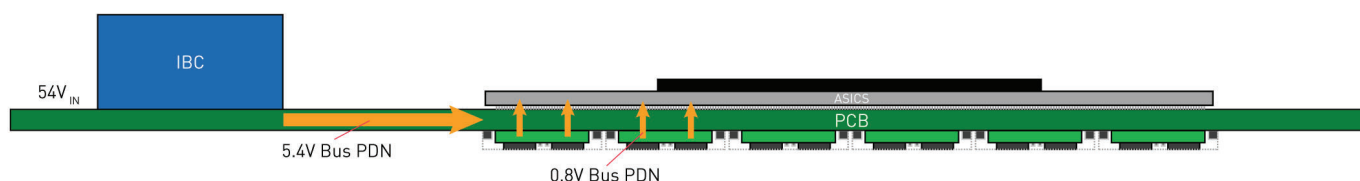
Today's datacenters use a lateral power delivery system, in which the power voltage regulators are placed on the top side of the board around the processor. As the current demand of CPUs and GPUs increases, the distance between the voltage regulator and the point of load become a significant factor for PDN losses. In addition, increased PDN means that the system requires a large amount of output capacitance for the voltage regulators.

Z-Axis Power Delivery™ is a method by which the voltage regulators are placed on the bottom side of the PCB below the processors. This results in a significant reduction (more than 10x) in PDN losses.

Lateral Power Delivery <1000A



Z-Axis Power Delivery™ >1000A



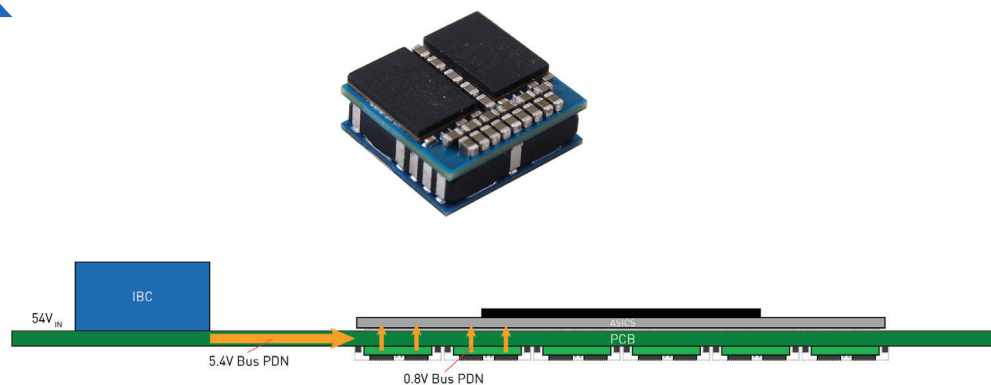
MPS offers power module solutions with <4mm height to fit in the limited space on the bottom side of the PCB.

Contact MPS for more information on Z-Axis Power Delivery™ EVBs

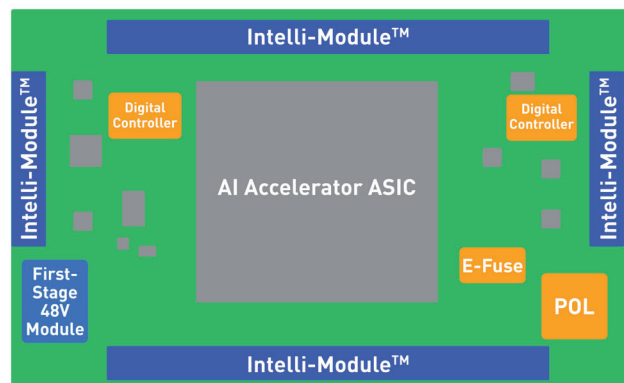
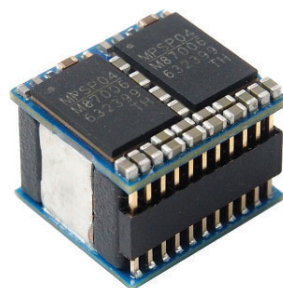
Adapting Power Solutions to the Increased Current Requirement of AI Processors

Increased Current Requirements

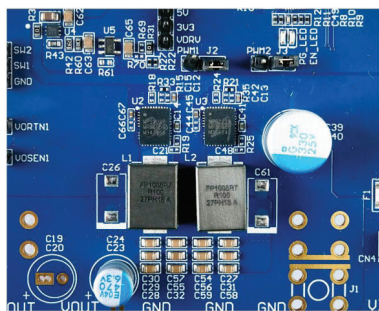
Z-Axis Power Delivery™
Ex: MPC22166-A



Lateral Power Delivery Using Modules
Ex: MPC22167



Lateral Power Delivery Using DrMOS
Ex: MP87006



First-Stage Solution: Power Modules

MPS offers multiple options for first-stage modules

8:1 & 10:1 IBC Modules

54V to 6.75V or 5.4V

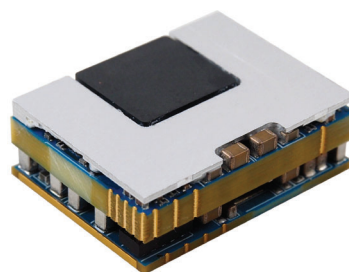


Enables a higher-frequency second stage due to lower switching losses, which allows the use of a smaller, shorter inductor

- Best for Z-Axis Power Delivery™ (ZPD)

4:1 IBC Modules

54V to 13.5V



Higher power density, established common footprint in the market

- Best for lateral power delivery (LPD)

Regulated Modules

54V to 12V



Targeted for systems such as CPU servers that require a regulated 12V supply

All of MPS's IBC modules can be paralleled to reach >1kW power levels.

MPC1100C-54 – 10:1 Fixed-Ratio, Digital, DC/DC 300W Power Module

The MPC1100C-54 is an open-frame, digital DC/DC power module with continuous power up to 300W. This module accepts up to a 60V input, and is configured in a 10:1 input-to-output ratio. It offers a digital controller with a multiple-time programmable (MTP) memory, which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

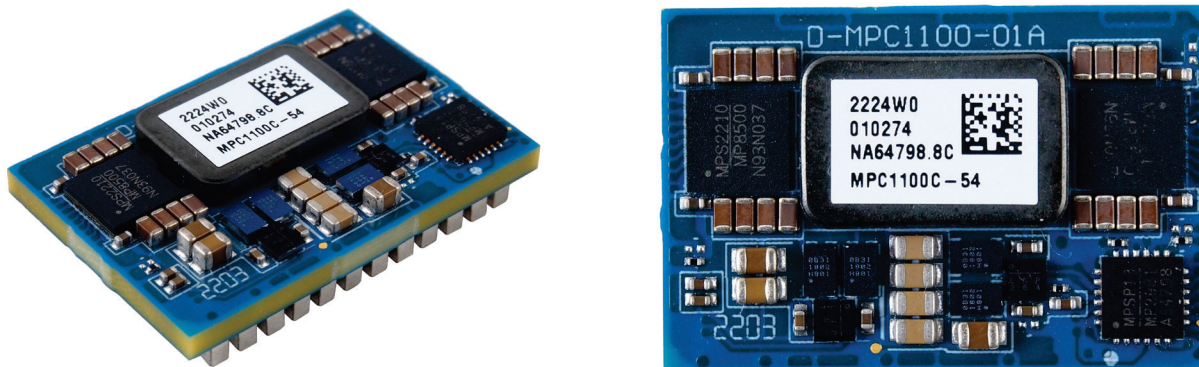


Figure 3: MPC1100C-54

Features:

- Input Voltage Range: 40V to 60V
- 10:1 Input-to-Output Ratio
- Power Density: 1700W/in³
- Non-Isolated LLC Topology
- Compatible with Serial Interface
- Supports Parallel Operation
- Built-In MTP to Store Custom Configurations
 - Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided

Dimensions:

27mmx18mmx6.1mm

Part Number	V_{IN} (V)	V_{OUT} Range (V)	Power Density (W/in ³)	P_{OUT} (W)	Peak Efficiency (%)	Efficiency @ 300W (%)
MPC1100C-54-0002	40 to 60	4 to 6	1700	300	96.7	95.3

MPC10106-54-0500 – 8:1 Fixed-Ratio, Digital, DC/DC 500W Power Module

The MPC10106-54-0500 is an open-frame, digital DC/DC power module with continuous power up to 500W. This module accepts up to a 60V input, and is configured in an 8:1 input-to-output ratio. It offers a digital controller with a multiple-time programmable (MTP) memory, which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

Features:

- Input Voltage Range: 40V to 60V
- 8:1 Input-to-Output Ratio
- Power Density: 2634W/in³
- Non-Isolated LLC Topology
- Compatible with Serial Interface
- Built-In MTP to Store Custom Configurations
 - V_{IN} Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided

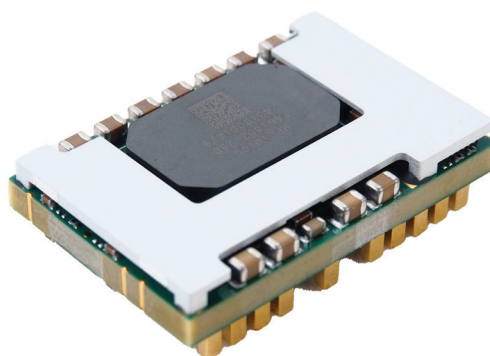


Figure 4: MPC12106-54-0750

Dimensions:

24mmx18mmx6.4mm

Part Number	V_{IN} (V)	V_{OUT} Range (V)	Power Density (W/in ³)	P_{OUT} (W)	Peak Efficiency (%)	Efficiency @ 750W (%)
MPC10106-54-0500	40 to 60	5 to 7.5	2634	500	97.8	95.9

MPC12106-54-0750 – 4:1 Fixed-Ratio, Digital, DC/DC 800W Power Module

The MPC12106-54-0750 is an open-frame, digital DC/DC power module with continuous power up to 750W. This module accepts up to a 60V input, and is configured in a 4:1 input-to-output ratio. It offers a digital controller with a multiple-time programmable (MTP) memory, which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

Features:

- Input Voltage Range: 40V to 60V
- 4:1 Input-to-Output Ratio
- Power Density: 3310W/in³
- Non-Isolated LLC Topology
- Compatible with Serial Interface
- Built-In MTP to Store Custom Configurations
 - V_{IN} Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided

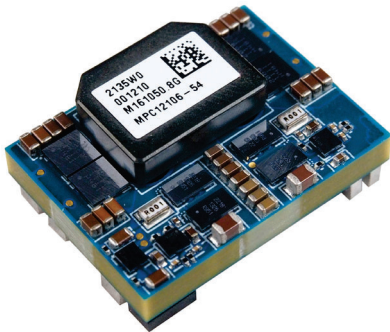


Figure 5: MPC12106-54-0750

Dimensions:

24mmx18mmx8.6mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	Power Density (W/in ³)	P _{OUT} (W)	Peak Efficiency (%)	Efficiency @ 750W (%)
MPC12106-54-0750-xxxx	40 to 60	10 to 15	3310	800	98.1	96.5

MPC42013 – Regulated, 12V, Digital DC/DC 1300W and 1600W Power Modules

The MPC42013 is a quarter-brick, open-frame, digital DC/DC power module with continuous power up to 1300W and 1600W. This module operates from a 40V to 60V DC primary bus to a regulated 12V output voltage. It offers a digital controller with a multiple-time programmable (MTP) memory, which can be easily configured to quickly bring up the system and nimbly adapt to system requirements.

Features:

- Input Voltage Range: 40V to 60V
- Power Density: 906W/in³
- Hybrid Buck Topology
- Compatible with Serial Interface
- Option for Active Current-Sharing in Parallel or Differential Remote Sense Mode
- Built-In MTP to Store Custom Configurations
 - V_{IN} Under-Voltage Lockout (UVLO)
 - V_{OUT} Over-Voltage Protection (OVP) and Under-Voltage Protection (UVP)
 - OCP_TDC/SPIKE
 - Over-Temperature Protection (OTP)
- GUI Configuration Software Provided



Figure 6: MPC42013

Dimensions:

58.4mmx36.8mmx13.2mm

Part Number	V_{IN} (V)	V_{OUT} (V)	Power Density (W/in ³)	P_{OUT} (W)	Peak Efficiency (%)	Efficiency @ 750W (%)
MPC42013-54-1300	40 to 60	12	751	1300	97.8	97.4
MPC42013-54-1600	40 to 60	12	924	1600	97.8	97

Second-Stage Solution: Digital Multi-Phase Controller + Intelli-Phase™ or Intelli-Module™

MPS has a robust product portfolio of digital multi-phase controllers and Intelli-Phase™ and Intelli-Module™ solutions.
For more information, visit monolithicpower.com/en/products/power-management/data-center.html

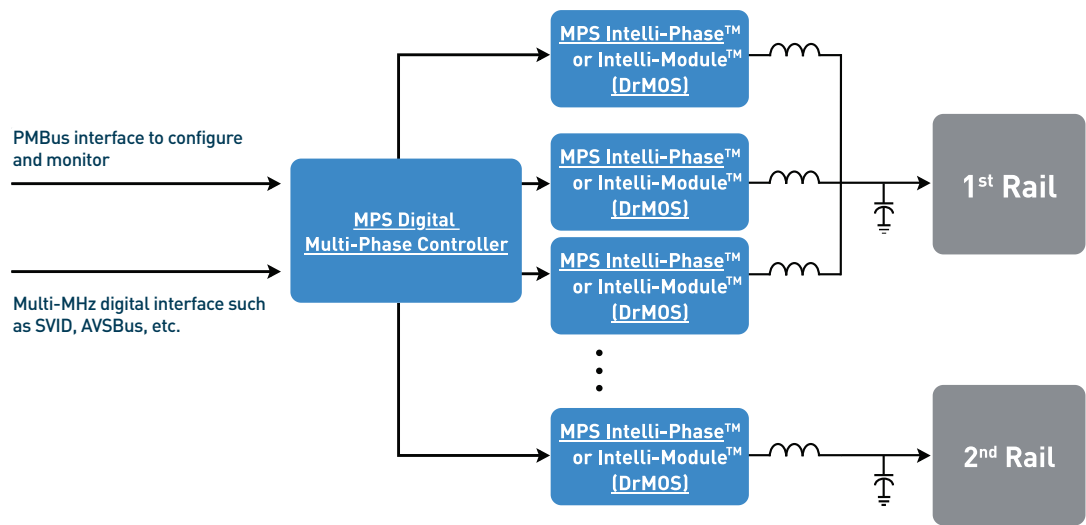


Figure 7: Second-Stage Digital Multi-Phase Controller + Intelli-Phase™ or Intelli-Module™ Block Diagram

Digital Multi-Phase Controller Features:	Intelli-Phase™ Features:	Intelli-Module™ Features:
- Patented Constant-On-Time (COT) Control - Multiple Loops, Flexible Phase Assignment - Interface Options: SVID, SVI2/3, OVR, or AVSBus - Fault Handling and Telemetry Capabilities	- Family of Devices Ranges from 12A to 100A - Wide f_{sw} Range: Balance between Transient and Efficiency - Built-In ZCD, OCP, NOCP, OTP, and SCP - Minimal External Components	- High Power Density in Space-Constrained Applications - Flexible Output Assignment - Support Parallel Operation for Higher Power Delivery - Optimized for Top-Side Cooling

Type of Product	Part Number	Part Description
Digital multi-phase controller	MP2975	Dual loop, 8-phase digital multi-phase controller
	MP29816	Dual-loop, 16-phase digital multi-phase controller
	MP2928	Triple loop, 12-phase digital multi-phase controller
DrMOS Intelli-Module™	MP87006	16V, 90A Intelli-Phase™ in 4mmx6mm package
	MP87001-A	16V, 90A Intelli-Phase™ in 5mmx6mm package
	MPC22157	16V, 130A Intelli-Module™ in 9mmx10mmx8mm footprint
	MPC22165A	16V, 170A Intelli-Module™ in 9mmx10mmx4mm footprint

MPC22166-A – 130A, Two-Phase Intelli-Module™ with Quiet Switcher™ Technology

The MPC22166-A is a non-isolated, step-down power module with 130A of continuous output peak current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 4mm maximum height makes it suitable for many applications, such as Z-Axis Power Delivery™ and PCIe form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Top-Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

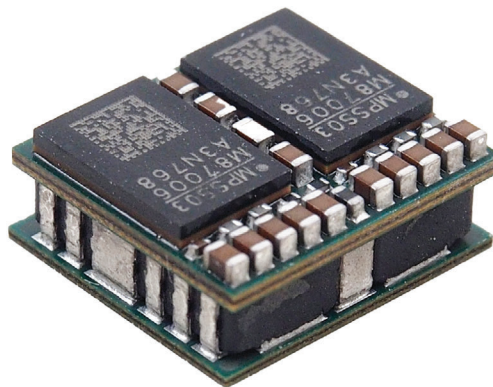


Figure 7: MPC22166-A -130A

Dimensions:

9mmx9.9mmx3.9mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{SW} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22166-A-130	4 to 16	0.5 to 2	130	600 to 1200	4904	89

MPC22166-B – 130A, Two-Phase Intelli-Module™ with Quiet Switcher™ Technology

The MPC22166-B is a non-isolated, step-down power module with 130A of continuous output peak current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 4mm maximum height makes it suitable for many applications, such as Z-Axis Power Delivery™ and PCIe form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.4V to 2V
- 130A Output Current
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

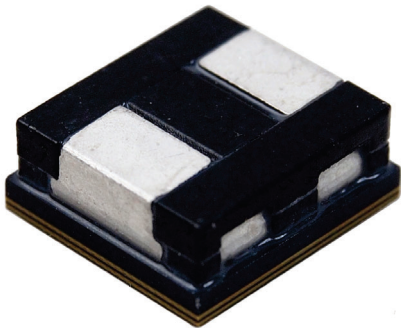


Figure 8: MPC22166-B -130A

Dimensions:

9mmx9.9mmx3.9mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22166-B-130	4 to 16	0.4 to 16	130	500 to 1500	4904	91

MPC22167 – 130A, Two-Phase, Intelli-Module™ with Quiet Switcher™ Technology

The MPC22167 is a non-isolated, step-down power module with 130A of continuous peak output current. This module integrates driver MOSFETs and an inductor in a compact package to save board space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 8mm maximum height makes it suitable for a wide range of applications, such as OAM form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Top-Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

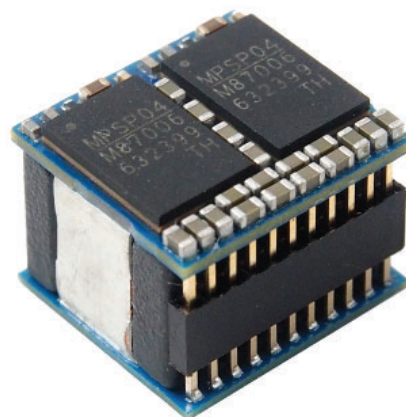


Figure 9: MPC22167

Dimensions:

9mmx9.9mmx7.65mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency @ 12V (%)
MPC22167-130	4 to 16	0.5 to 2	130	400 to 1500	2475	91

MPC22157 – 130A, Two-Phase Intelli-Module™ with Quiet Switcher™ Technology

The MPC22157 is a non-isolated, step-down power module with 130A of continuous output peak current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 8mm maximum height makes it suitable for many applications, such as OAM form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 130A Output Current
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package



Figure 10: MPC22157 -130A

Dimensions:
9mmx9.9mmx7.7mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22157-130	4 to 16	0.5 to 2	130	500 to 1500	2484	92

MPC22165-A – 170A, Two-Phase Intelli-Module™ with Quiet Switcher™ Technology

The MPC22165-A is a non-isolated, step-down power module with 170A of continuous output peak current. This module integrates driver MOSFETs and an inductor in a compact package to save layout space and achieve a higher power density. It is scalable for many modules in parallel, up to 2kW or more of power. The 10mm maximum height makes it suitable for many applications, such as OAM form factor boards.

Features:

- Input Voltage Range: 4V to 16V
- Output Voltage Range: 0.5V to 2V
- 170A Output Current
- Top Side Cooling Package
- Current Sense: Accu-Sense™
- Temperature Sense
- Current Limit Protection
- Supports Parallel Operation
- Over-Temperature Protection (OTP)
- Fault Reporting
- Common Footprint LGA-72 (9mmx10mm) Package

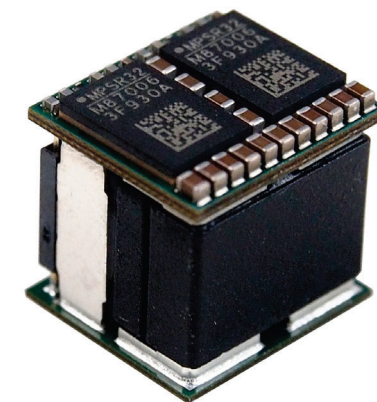


Figure 11: MPC22165-A -170A

Dimensions:

9mmx9.9mmx9.7mm

Part Number	V _{IN} (V)	V _{OUT} Range (V)	I _{OUT} (A)	f _{sw} (kHz)	Power Density (W/in ³)	Peak Efficiency (%)
MPC22165-A-130	4 to 16	0.5 to 2	170	600 to 1500	2579	90

MP29816 – Dual-Loop, AVSBus and OVR-16, 16-Phase Controller

Features:

- Highest Phase Count
 - Dual-Loop, 16-Phase Controller Provides Power for Nvidia PWM-VID and AVSBus Core Power
 - Stackable Controller Supports 32, 48, or 96 Phases, and Improves Current and Thermal Balance Compared to 1 PWM Driving 2 DrMOS
- Digital Control
 - Digital Loop Compensation and Load-Line Regulation
 - Nonlinear Control Reduces Output Capacitor Requirements for Transients
- Digital Configurability and Monitoring
 - Output Voltage/Current, Input Voltage/ Power and Temperature Telemetry
 - Over-Power (OP) Limit and Protection
 - Intelli-Phase™ Fault Diagnosis
 - Integrated Non-Volatile Memory (NVM) to Store VR Programming and for Black Box Capability
 - TLVR Fault Detection
- Design Flexibility
 - Automatic Phase-Shedding (APS) Improves Light-Load Efficiency
 - Configurable Voltage, Current, and Temperature Protections



Package:
TQFN-56 (7mmx7mm)

MPQ8785 – 16V, 40A, Scalable, Digital, Synchronous Step-Down Converter with PMBus

Features:

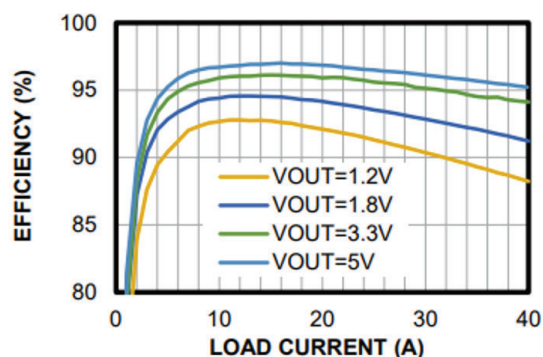
- V_{IN} Voltage Range
 - 3.1V to 16V with External 3.3V VCC Bias
- Output Current & Voltage
 - Continuous Output Current: 40A
 - Stackable (Multi-Phase) to Support 640A (16 POLs in Parallel)
 - Adjustable Output Voltage (V_{OUT}) from 0.35V to $0.9 \times V_{IN}$, Up to 5.5V Max
- Buck Regulator
 - Adaptive Constant-On-Time (COT) Control
 - PMBus 1.3
 - Accurate Voltage, Current, and Telemetry Reporting
 - Accurate Current-Balance Control in Multi-Phase Configuration
 - Output Voltage True Remote Sense
- Additional Features
 - Precision Reference Voltage (V_{REF}), $\pm 0.5\%$ over 0°C to 70°C Junction Temp Range
 - 300kHz to 2MHz Configurable f_{SW}
 - Quiet Switcher™ Technology
 - Fault Recording (Black Box) to Ease Failure Analysis
 - Phase Redundancy
 - Automatic PMBus Address Assignment to Provide Auxiliary Phase Reading
- Protections
 - Configurable Hiccup or Latch-Off Mode
 - Input Voltage (V_{IN}) Under-Voltage Protection (UVP) and Over-Voltage Protection (OVP), V_{OUT} UVP and OVP
 - Adjustable Output Current Limit
 - Thermal Warning and Shutdown



Package:
TLGA-37 (5mmx6mm)

Efficiency vs. Load Current

1 phase CCM, $F_{sw}=800\text{kHz}$,
 $L=150\text{nH}/0.17\text{m}\Omega$



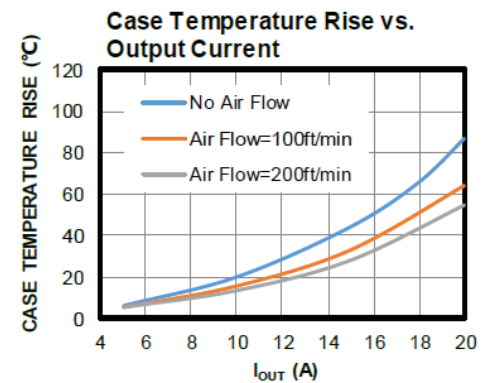
MP5048 – 60V, 15A, 7mΩ $R_{DS(ON)}$ Hot-Swap Intelli-Fuse Solution

Features:

- 60V E-Fuse with Highest Industry Current
- Flexible for Multiple Phases in System
 - Input Voltage Range: 24V to 60V
 - Scalable for Standalone or Parallel Operation
 - Configurable Output Voltage Soft Start Supports Inrush Current Control
 - 1.5% IMON Reporting Accuracy Supports Accurate System Power Reporting
- Full System Protections
 - Configurable Over-Current Protection (OCP) Limit
 - Accurate On-Die Temperature Sensing and Thermal Shutdown
 - Pin-Configurable Latch-Off and Hiccup Protections
 - Fault Pin Indicator (FLTBI)



Package:
QFN-30 (5mmx5mm)



Quality Assurance & Reliability Commitment

The MPS Quality Assurance organization develops, coordinates, and champions strategic quality initiatives throughout MPS Inc., its foundries, and subcontractors. Its mission is to enable MPS to design, develop, manufacture, and deliver products to our customers with world-class quality and reliability that meet and exceed our customers' expectations.

MPS and Its Supplier Quality Systems and Certificates:

- ISO 9001:2008 (MPS)
- EU RoHS/HF/REACH Compliant (MPS)
- Sony Green Partner (MPS & Suppliers)
- TS16949 (Suppliers)
- ISO 14001 (Suppliers)

Product Quality:

- Automotive Products Qualified per AEC-Q100 Standard
- Standard Products Qualified per JEDEC and Military Standard
- Reliability Failure Rate <10FIT
- Product Quality Level <1.0ppm

Quality Control and Monitor:

- On-Site Foundry and Assembly Teams for Real-Time Actions
- Quarterly Supplier Quality Review and Annual Supplier Audit
- Short-Term Reliability Monitor Test – Daily
- Long-Term Reliability Monitor Test – Monthly
- Real-Time Engineering Actions on Monitor Failure
- Quarterly Reliability Monitor Reports

About MONOLITHIC POWER SYSTEMS

Who we are

We are creative thinkers. We break boundaries. We take technology to new levels. As a leading international semiconductor company, Monolithic Power Systems (MPS) creates cutting-edge solutions to improve the quality of life with green, easy-to-use products.

What we do

We make power design fun! With our innovative proprietary technology processes, we thrive on reimagining and redefining the possibilities of high-performance power solutions in industrial applications, telecom infrastructures, cloud computing, automotive, and consumer applications.

Where we come from

It started with a vision. Michael Hsing, pioneering engineer and CEO, founded Monolithic Power Systems, Inc. in 1997 with the belief that an entire power system could be integrated onto a single chip. Under his leadership, MPS has succeeded not only in developing a monolithic power module that truly integrates an entire power system in a single package, but also it continues to defy industry expectations with its patented groundbreaking technologies.

Our values

We cultivate creativity

As a company we believe in creating an environment that encourages and challenges our employees to collaborate and think outside the box to excel beyond their preconceived capabilities.

We do not accept the status quo

We do not believe in limitations. It is not about what is, but what can be. Possibilities are endless at MPS.

We are passionate about sustainability

It's about the future. From materials to finances, we are committed to conservation. We will not tolerate waste in an effort to improve and preserve the quality of life.

We are committed to providing innovative products to our customers

Let us do the heavy lifting. We relentlessly strive to make system design versatile and effortless to meet our customers' specific needs. We'll do the work, so our customers can have the fun!

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MPS 48V Datacenter Solutions

DC/DC Power Conversion for Datacenter,
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Vertical Power Delivery for 1000 Amps Machine Learning ASICs

Houle Gan, Shuai Jiang, Sue Teng, Shin Yamamoto, Venkata Chivukula, Bill Edwards, Chee Chung, Jason Chen, Mushafik Mohideen, Gregory Sizikov, Xin Li

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Sunnyvale, USA
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Abstract—The recent explosive growth of machine learning (ML) applications has resulted in a surge of demand for compute performance from custom ASICs, and with the ending of Dennard scaling, power demands of such ASICs have also reached unprecedented levels. In this paper we describe a 48V to point-of-load power solution capable of delivering more than 1000 Amps of current at 0.8V for the latest ML ASICs. The solution follows a two-stage approach, with a fixed-ratio high-density high-efficiency first stage and a second stage module with very high density ($1\text{A}/\text{mm}^2$, $1300\text{W}/\text{inch}^3$). The second stage module is integrated into the system board by surface mounting directly underneath the ASIC. This Vertical Power (VPWR) solution removes the “last mile” power loss found in the conventional lateral design, lowers I^2R loss by 70% for 1000 Amps load current, and enables many more other benefits for the overall system such as better signal integrity and thermal performance.

Keywords—vertical power delivery, high density, voltage regulator modules (VRM), machine learning ASICs, 1000 Amps

I. INTRODUCTION

With years of development, the facility level and rack level power solutions of modern data centers are highly efficient, with PUEs as low as 1.1 [1]. However on the tray level PCB board, from the point-of-load (PoL) voltage regulator (VR) to the ASIC load, power is converted to low voltages with very high current and travels for a final lateral distance of a few tens of millimeters with heavily penetrated metal planes, and that’s where the majority of power loss happens, as shown in Fig. 1. This IR loss is proportional to the square of current, hence grows much faster than ASIC’s own power demand which is only linearly proportional to current. For the latest ML ASICs [2, 3] with lateral power solutions, the distribution power loss alone is estimated to be 5-20% of the ASIC’s thermal design power (TDP). This percentage is expected to further increase as ASIC’s power scales up. Vertical Power Delivery (VPWR) has been an emerging solution for this lossy “last mile” of lateral distribution [4], as shown in Fig. 2. In the following sections, we present a VPWR solution to convert and deliver more than 1000 Amps of current from directly underneath the ASIC, with $1\text{A}/\text{mm}^2$ output current density for the vertical module and 70% power loss reduction. With the introduction of the solution, the distribution power loss was reduced from more than 20% of ASIC TDP to below 7%, as shown in Fig. 3.

To authors’ best knowledge, this is the first report of a mass-produced, high-density, high-efficiency Vertical Power solution with 1000 Amps output current for ASICs that are not wafer scale packaged.

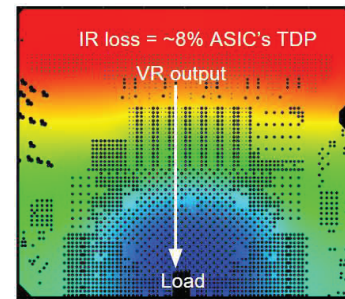


Fig. 1. Distribution loss in lateral power solutions.

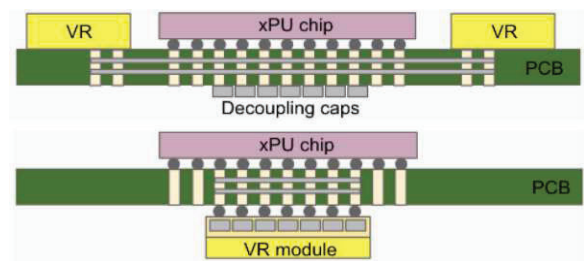


Fig. 2. Lateral power vs. VPWR solutions.

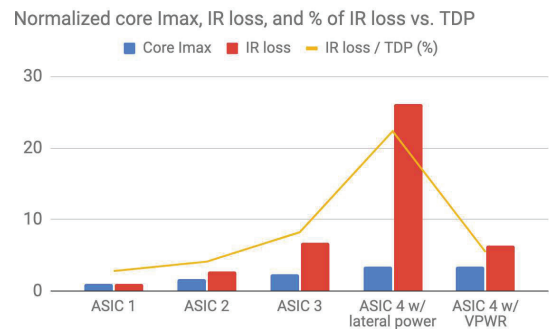


Fig. 3. Distribution power loss vs. chip TDP.

II. OVERALL ELECTRICAL DESIGN

The overall power design architecture follows a two-stage approach [5], with high-efficiency and high-density fixed-ratio converters as the first stage converting 48V tray input power to an intermediate bus voltage, and followed by a second multi-phase Buck stage that converts further down to load voltage range of 0.6-1V, as shown in Fig. 4.

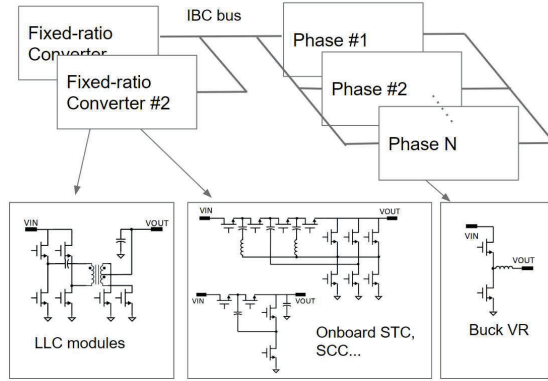


Fig. 4. Two-stage 48V-PoL VPWR architecture.

The intermediate bus voltage level was carefully chosen for the optimal efficiency for the end-to-end system. Key considerations included first-stage efficiency, second-stage power stage efficiency at different input voltages, and the IR loss on the bus plane. Fixed ratios including 10:1 and 8:1 designs were evaluated, and the 8:1 IBC with a 6.75V nominal intermediate voltage was chosen for the final design. The switching frequency of the second stage Buck converter was also carefully studied, considering transient performance requirements, inductor / output caps sizing, and efficiency implications. A switching frequency of 1.2MHz was chosen for the final design. Other techniques such as coupled inductors and trans-inductor VR (TLVR) [6] could be utilized as well if better transient performance is required.

III. POWER MODULE CONSTRUCTION

The first stage IBC converters were placed laterally to the load ASIC, while the second stage VRM was mounted to the bottom of the PCB directly underneath the ML ASIC. A total of 24 phases of power stages were used in the final design, to support 1000 Amp of TDP and 1300 Amp of peak power. Inductors were embedded into the thick PCB underneath the phases. The PCB carrying all the power stages and the inductors was named “power board”. On the other hand, as indicated in Fig. 2, decoupling caps were originally distributed on the bottom side of the main board under the ASIC, to reduce the mid-frequency range power delivery network (PDN) impedance or to mitigate the so-called second droop. In the proposed design, these decoupling caps were absorbed into the second stage VPWR module as well, onto an additional PCB to the power board. This additional “cap board” was then bonded with the power board. The bottom side of the cap board was planted with solder ball arrays that match the ASIC package ball pattern, and the module was then ready to be surface mounted to the main board.

The final second-stage VRM module which is placed vertically, is shown in Fig. 5. The module size is $\sim 33 \times 38 \times 7.5 \text{ mm}$, with an output current density of 1 A/mm^2 and power density of 1300 W/inch^3 . Before being placed onto the production board, a test system has been built with a 1000W water-cooled custom load board, as shown in Fig.6. Extensive study has been conducted for not only electrical functionality but also mechanical and thermal properties of the module with custom test vehicles. Weight per ball, BGA ball material, compressive loading capability, warpage, thermal expansion coefficients are among the items that were examined in detail. Reliability and environmental quality testing has also been conducted as well to demonstrate mass production readiness.

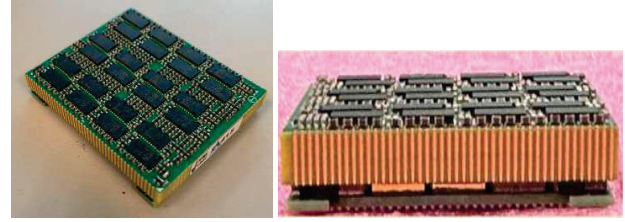


Fig. 5. VPWR module and test board. (a) Top view. (b) Side view.

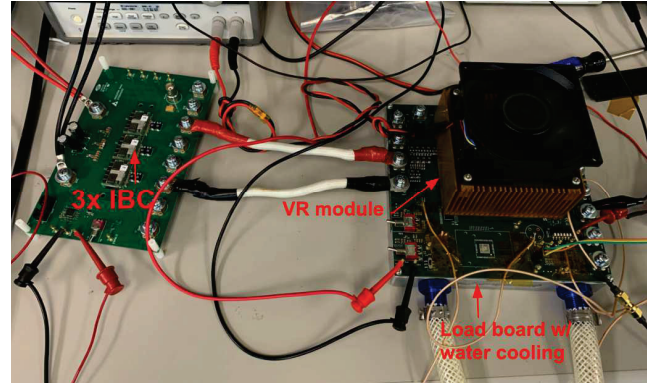


Fig. 6. VPWR test system with custom load board and water cooling.

IV. SYSTEM INTEGRATION

An illustration of the VPWR module’s integration with the ASIC and system board is depicted in Fig. 7, and a photo of assembled VPWR modules with the bottom side cooling assembly is shown in Fig. 8. The VPWR module interacts with the rest of the system in very different ways from the conventional lateral power design. Mechanically the VPWR module impacts the ASIC package and PCB warpage during and after reflow, changing the surface flatness contours of both top and bottom sides. This effect is examined through numerical simulation and correlated with Shadow Moire measurements, as shown in Fig. 9. Thermally, the VPWR module adds one substantial heat source to the vertical stack, but the IR loss underneath the ASIC area is also much lower than lateral power design. With the addition of a bottom side heatsink, it is shown that the ASIC BGA temperature is actually lower than the lateral design. Lower BGA temperature greatly improves the solder ball’s electro-migration capacity,

as simulated in Fig. 10, which is a critical limitation in ultra-high power designs.

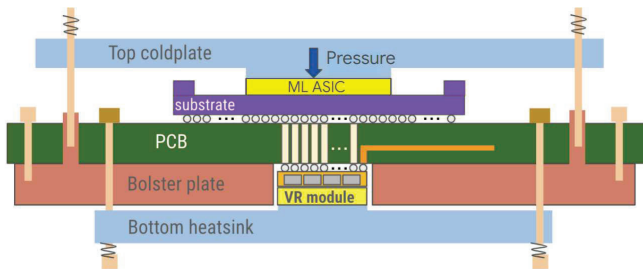


Fig. 7. System integration of VPWR module with ASIC and system board. (a) Side view. (b) Exploded view.

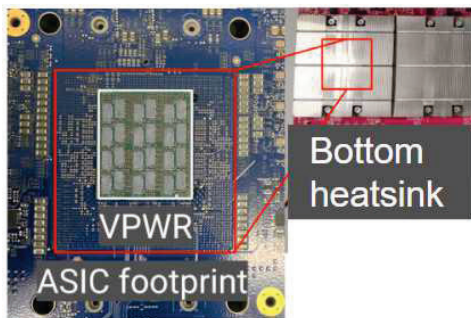


Fig. 8. Assembled VPWR solution with bottom heatsink.

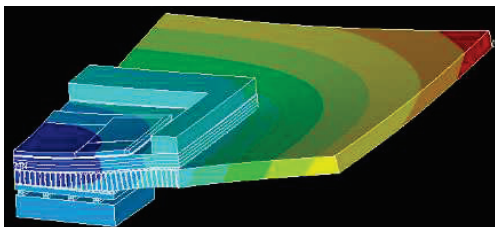


Fig. 9. Structural analysis of VPWR, PCB and ASIC combined.

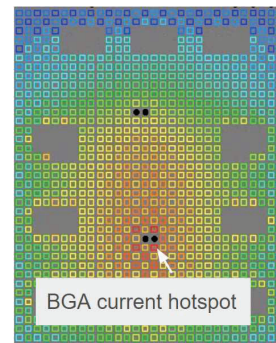


Fig. 10. BGA current heatmap to evaluate solder ball electro-migration risks.

For manufacturability, the dual side SMT process required careful development. The mixture of via-in-pad (VIPPO) and non-VIPPO vias, combined with the large thermal mass of the VPWR module itself, could easily result in solder hot tear defects on both the ASIC and VPWR module BGA interfaces [7], as shown in Fig. 11. Reworking such modules is challenging, requiring non-standard processes with custom nozzles, compression fixtures, and heat shields, which demonstrated successes in mass production. Finally, the VPWR solution reduced PCB size by roughly 10%, and opened up precious chip peripherals for high-speed SerDes routings. A study has shown that with VPWR solution, the lengths of high-speed PAM4 SerDes channels have been substantially reduced, enabling the use of lower cost fab laminates. The signal routings are also further away from noisy VR output nodes, benefiting high-speed signal integrity in general.

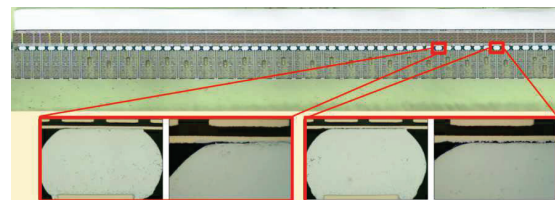


Fig. 11. Solder ball hot tear defects on VPWR and ASIC BGAs.

V. MEASUREMENT RESULTS

Power loss, stability, efficiency, and transient performances are simulated and measured on both the test systems and production boards. The load current was generated either by the custom load boards on the test system, or with realistic and synthetic workloads running on the real ML ASIC silicon on the production board. The measurement confirmed IR loss reduction from more than 100W to about 30W for 1000 Amp output current, indicating 70% reduction in IR loss. The 6.75V-to-1V VPWR module efficiency and the overall 48V-to-PoL efficiency curves are plotted in Fig. 12. Note that the latter curve includes all losses on the system board from tray input power to ASIC load, including first and second stages conversion loss, intermediate bus IR loss, output network IR loss, gate driving loss, and everything else. Bode plot measurement was taken with 1000 Amp load current and is shown in Fig. 13, with 68 degree of phase margin and 13.5dB of gain margin. Transient performance was measured with an

800A load step on both test systems with load boards and production boards with real ASICs. Undershoot was 60mV and 68mV respectively, and overshoot was 70mV and 92mV respectively. Note that the load step with the real ML ASIC and workloads is much faster than the bandwidth of the VR, so the measured performance reflects the performance of the full PDN and VR combined, as shown in Fig. 14.

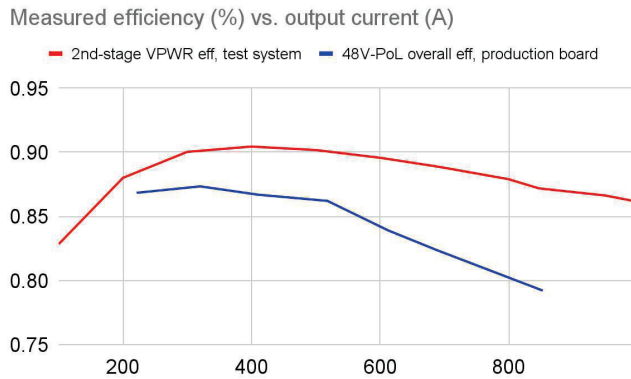


Fig. 12. Measured VPWR efficiency and overall 48V-to-PoL efficiency.

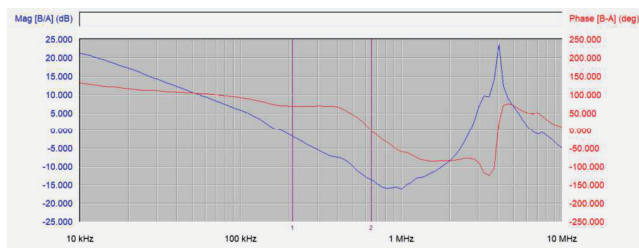


Fig. 13. Measured Bode plot of production boards.

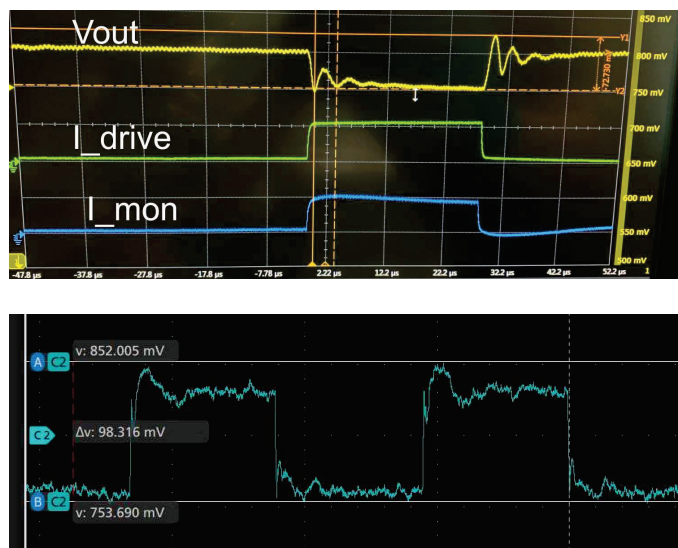


Fig. 14. Measured transient performance of 800A load steps on (a) test boards with custom load board, and (b) production boards with ASICs.

VI. CONCLUSIONS AND FUTURE WORK

In this paper we presented for the first time a mass-produced, high-density, high-efficiency Vertical Power solution with 1000 Amp output current, which combines many of state-of-the-art design, manufacturing and assembly technologies to enable ultra-high power ML ASICs. Our study has shown that ASICs with more than 600A current could reach the inflection point where IR loss in the lateral distribution path exceeds 10% of the chip's TDP. We believe that other than the latest ML ASICs, many chips in other applications are also rapidly approaching this point, including network switches and high-performance CPU/GPU chips. VPWR solution is beneficial for all these cases, not only by substantially reducing the distribution power loss, but also by enabling better system floor plan, improving signal integrity channels, and better managing thermal performance. The challenges are to understand the mechanical interactions and reliability implications with the complex vertical stack, and to ensure sufficiently wide process windows for robust manufacturing and assembly processes.

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Vertical Power Delivery for 1000 Amps Machine Learning ASICs

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Chivukula, Bill Edwards, Chee Chung, Jason Chen, Mushafik
Mohideen, Gregory Sizikov, Xin Li

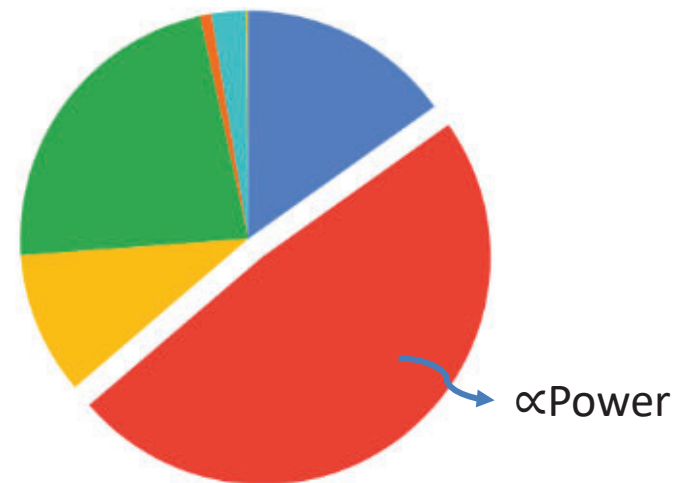
Google, LLC



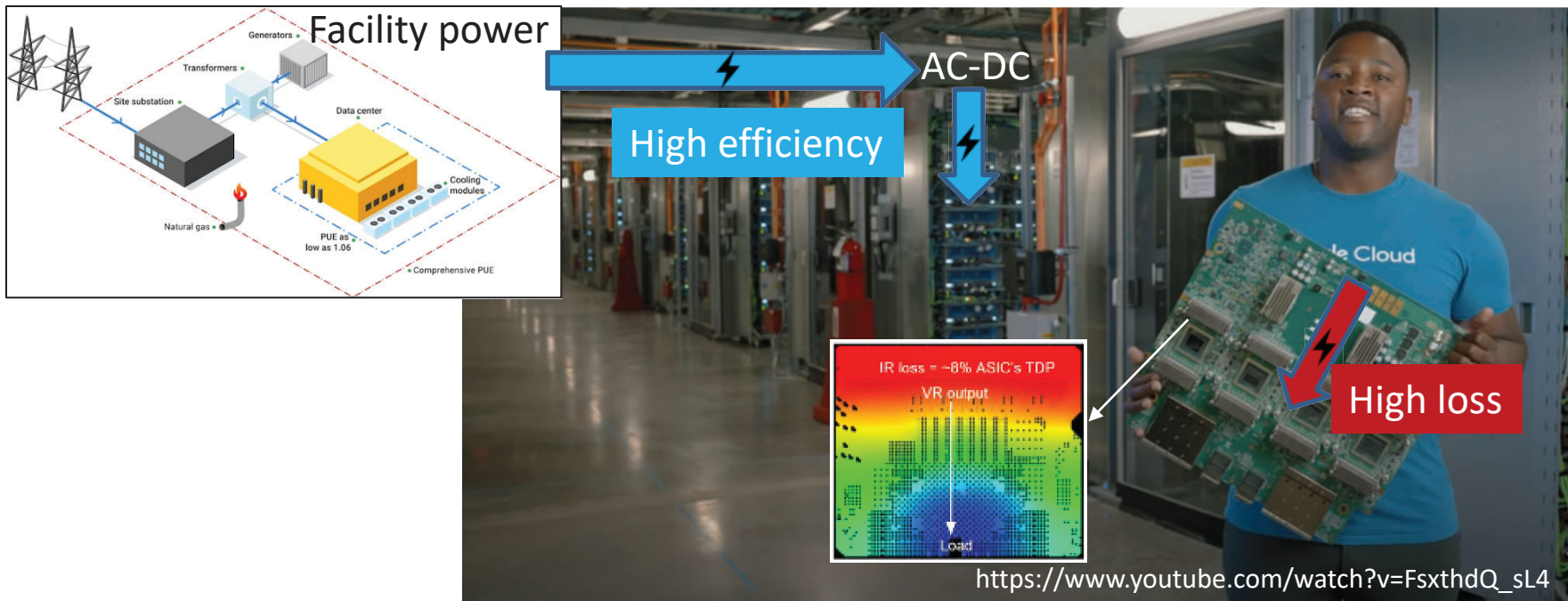
AI Infra Is Expensive



ML fleet TCO breakdown



Lossy “Last Mile” in Power Delivery

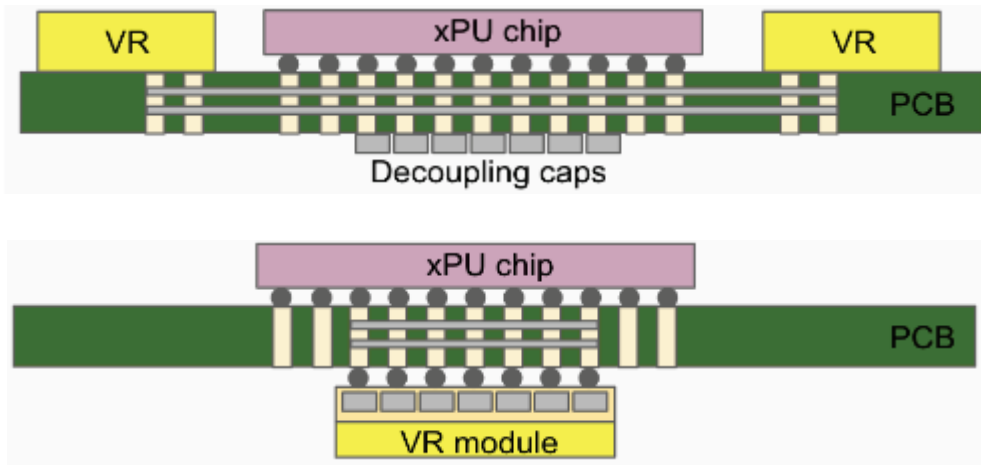


Lossy “last mile” - majority of power loss (sometimes >10% of chip TDP) is from the PCB/package

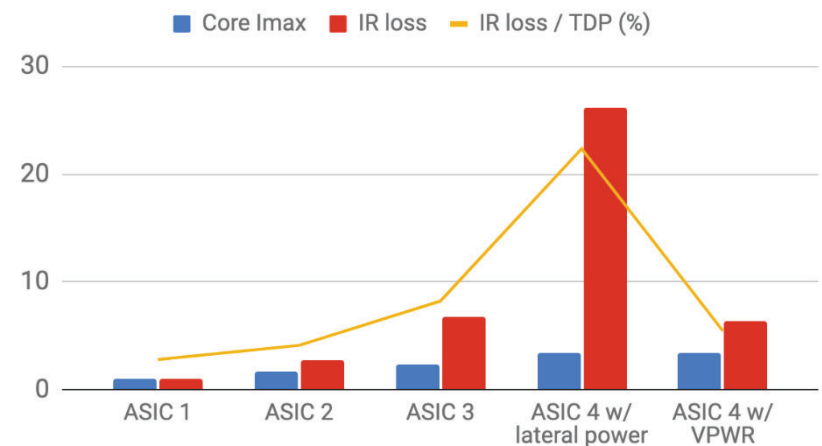
What is Vertical Power Delivery?

Vertical Power (VPWR):

- Delivering 1000 Amp to ASIC in z-dimension
- ~70% reduction in distribution power loss
- Many side benefits (e.g., substantially shorter SerDes channels)

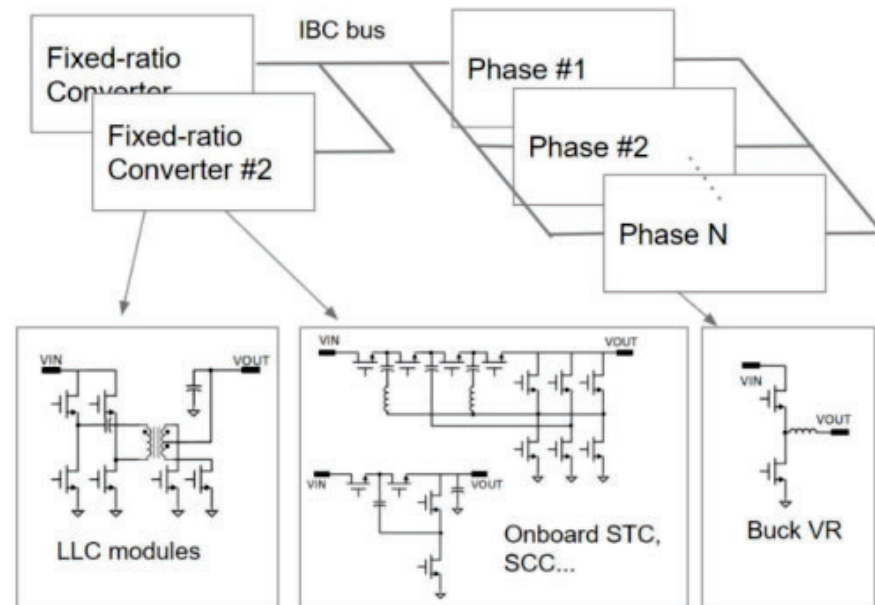


Normalized core I_{max}, IR loss, and % of IR loss vs. TDP



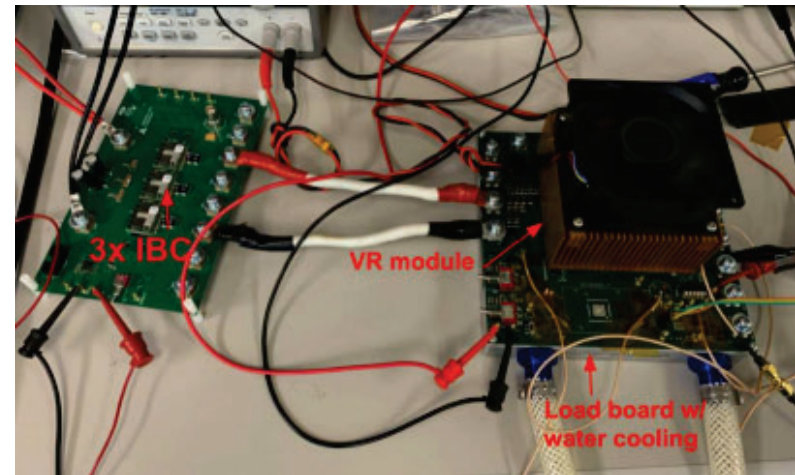
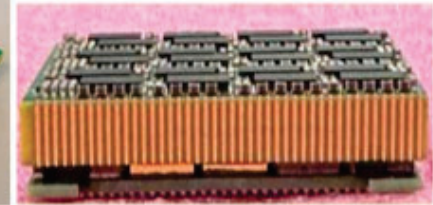
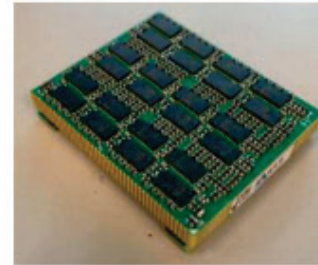
Electrical Design

- Two-stage 48V-PoL VPWR architecture
 - 8:1 IBC with 6.75V nominal intermediate voltage as first stage
 - Buck converter at 1.2MHz switching frequency as second stage
 - Coupled inductors and trans-inductor VR topologies compatible

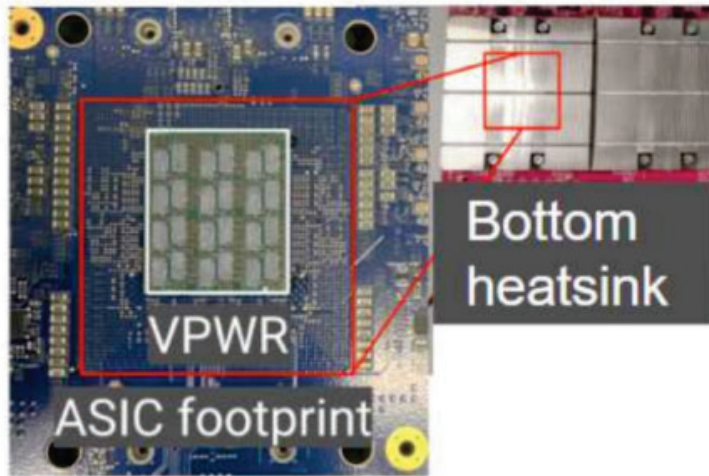


VPWR Module Construction

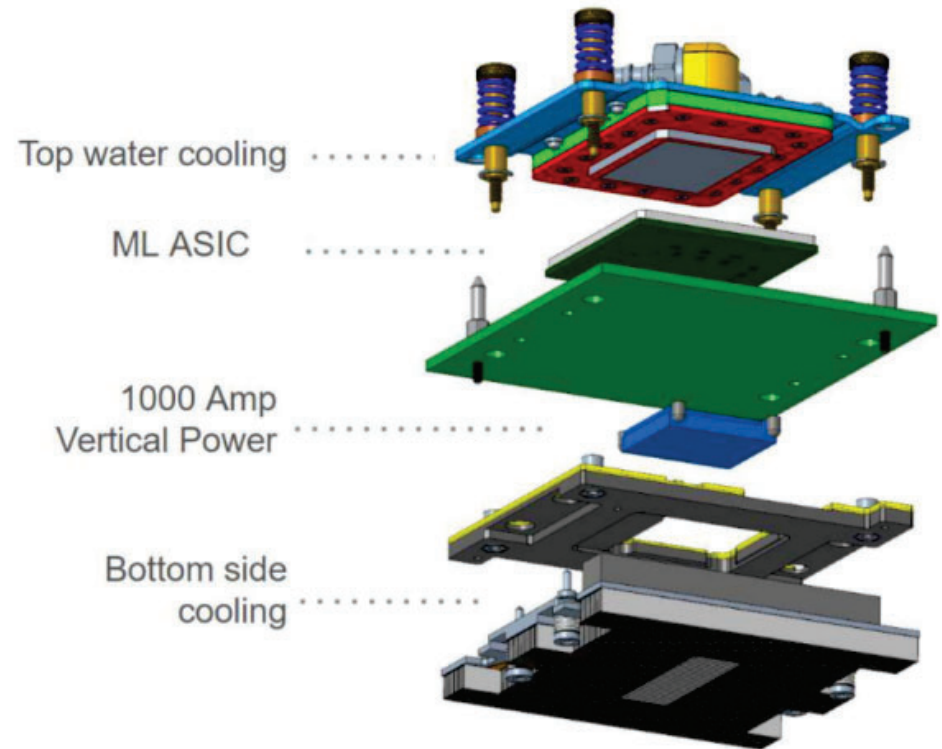
- Module size: $\sim 33 \times 38 \times 7.5 \text{ mm}$
- 1000A TDC, 1300A EDC
- High power density: 1 A/mm^2 , 1300 W/inch^3
- Dual-layer PCB structure
 - 24-phase power board
 - High-density cap board
- Test system with 1000W custom water cooled load board



System Integration

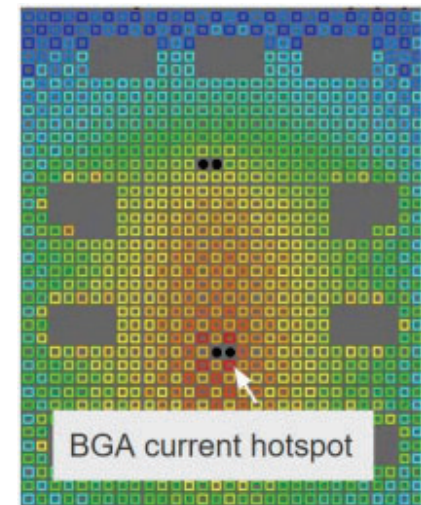
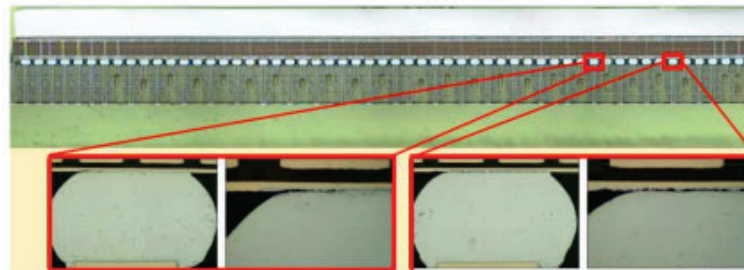
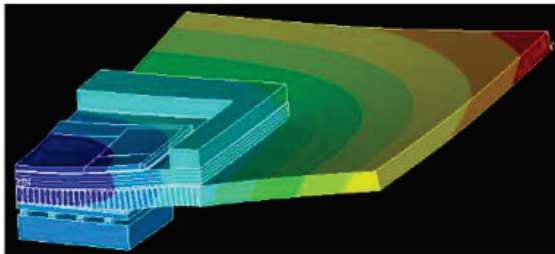


Picture of VPWR on production boards



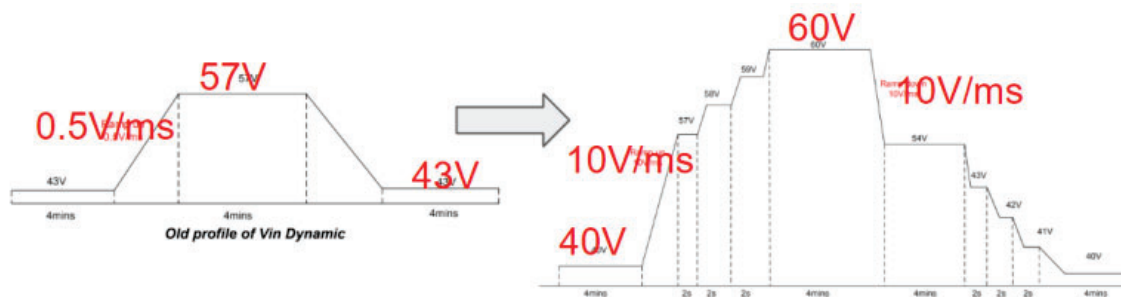
System Integration

- Unique challenges in all aspects of system integration
 - Mechanical design and structural integrity
 - Thermal impact
 - Power integrity, BGA EM
 - Dual side BGA assembly, weight, reworkability



Quality and Reliability

- Mass production ramp-up and readiness
 - Quality assurance on production line
 - Customized burn-in fixtures and systems
 - Much more stressful than typical power module burn-in
 - High slew-rate Vin/load shmoo, TCT, PMBus, etc.
 - Enhanced ORT (on-going reliability test)



Vin + Load sweep in burn-in chambers

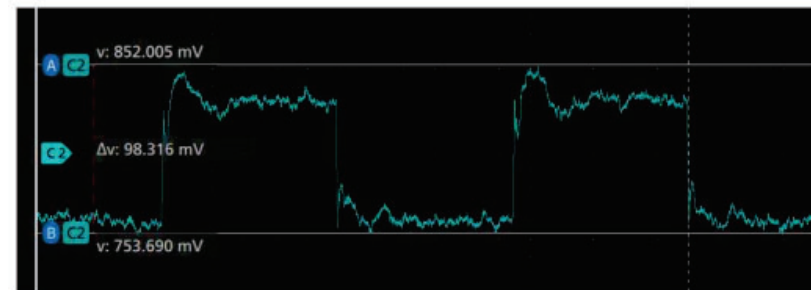
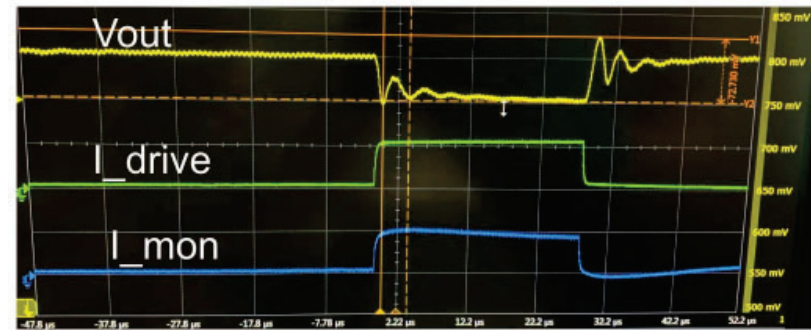
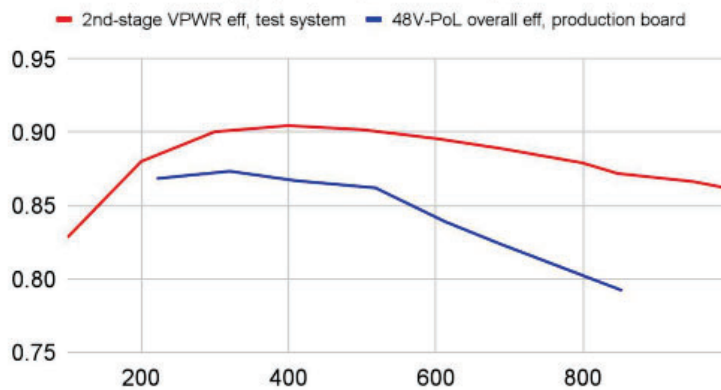


Custom ORT fixture

Measurement Results

- Industry's first 1000 Amps VPWR solution in volume production
 - 87% 48V-to-PoL overall peak efficiency
 - >80% 48V-to-PoL overall efficiency at 800A
 - 68mV undershoot with 800A step

Measured efficiency (%) vs. output current (A)



Vertical Power Roadmap

- First report of 1000-Amp VPWR solution in mass production
- Multiple generations of VPWR solutions in production or development
- Advanced module design
 - Multi-rail integration, controller integration
 - Advanced topologies such as TLVR for superior transient performance
 - Modular design and scalability
 - Advanced power management features
- Next-generation solution for ultra-high power ASICs
 - Multi-module design
 - Loss reduction on intermediate bus
- Enhancement in structural integrity, ease for manufacturing, and quality / reliability